

Graphical Abstract

Crowd Density Image Classification Framework: An Explainable CNN-Based Decision-Support System Using Transfer Learning and Grad-CAM for Public Safety Monitoring

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Highlights

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- CNN framework classifies crowd density into four levels for real-time public safety monitoring.
- Merged 7,558-image dataset from three public sources covers Low, Medium, High, and Very High density.
- EfficientNetB0 achieves 83.27% accuracy at 3.84 ms per image, outperforming ResNet50 and custom CNN.
- Grad-CAM confirms model attention focuses on crowd regions for correct density predictions.
- All three models achieve perfect Very High density classification, supporting reliable dangerous-crowd alerting.

Crowd Density Image Classification Framework: An Explainable CNN-Based Decision-Support System Using Transfer Learning and Grad-CAM for Public Safety Monitoring

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Abstract

Crowd-related disasters including the 2022 Itaewon crush and the 2024 Hathras stampede demonstrate that the failure to detect dangerous crowd density before escalation remains a critical and unresolved public safety challenge worldwide. Manual surveillance is reactive, cognitively demanding, and unable to provide continuous density monitoring across multiple camera feeds simultaneously at the speed required for early intervention. Existing deep learning studies on crowd analysis focus predominantly on regression-based head counting rather than discrete density classification, and no study in the reviewed literature combines a multi-source dataset, custom CNN versus multiple transfer learning comparison, class-imbalance handling, per-class evaluation, and inference time analysis within a single reproducible framework. This study addresses automated classification of crowd images into four density levels using a comparative and explainable CNN framework trained on a merged dataset of 7,558 images from ShanghaiTech, JHU-Crowd++, and Abnormal High-Density Crowds. Three models were implemented under identical conditions: a custom CNN baseline, ResNet50, and EfficientNetB0 using two-phase transfer learning, evaluated using accuracy, macro F1-score, ROC-AUC, and inference time. EfficientNetB0 achieved 83.27% test accuracy at 3.84 ms per image, followed by the custom CNN at 82.74% and ResNet50 at 81.48%, with all three models achieving perfect Very High density classification ($F1 = 1.00$). Grad-CAM visualisations confirmed that the custom CNN focuses on semantically meaningful crowd regions for correct predictions, with heatmap activation dispersing to background areas for misclassi-

fied samples. The proposed framework provides a reproducible, explainable decision-support pipeline for crowd safety monitoring, with SHAP analysis identified as the highest-priority future extension.

Keywords: Crowd Density Classification, Convolutional Neural Networks, Transfer Learning, Grad-CAM, Public Safety Monitoring, EfficientNetB0, ResNet50

1. Introduction

1.1. Application Background and Practical Importance

Crowd management at large public gatherings represents one of the most urgent and underserved challenges in public safety monitoring worldwide. The 2022 Itaewon Halloween festival crush in Seoul, South Korea, caused 158 deaths when a densely packed crowd became compressively asphyxiated in a narrow alleyway with no automated density warning in place [1]. The 2021 Astroworld music festival stampede in Houston, Texas, resulted in 10 fatalities when the crowd surged toward the performance stage without security personnel detecting the dangerous density level in time to respond [2]. The 2024 Hathras religious gathering stampede in Uttar Pradesh, India, claimed 121 lives when Very High density conditions developed faster than security operators monitoring manual feeds could detect and react to [3, 1]. Governments, event organizers, venue managers, and emergency services are all directly affected by the absence of reliable automated crowd density assessment systems, since human operator attention capacity is fundamentally limited across multi-camera surveillance environments [4]. Manual surveillance requires trained security personnel to visually monitor dozens of camera feeds simultaneously, introducing significant cognitive fatigue, delayed response times, and inconsistent density judgments between individual operators across long shifts [5]. The consequence of a missed or delayed density assessment in an overcrowded venue can escalate from discomfort to injury to mass casualty within minutes, leaving no practical margin for human-only monitoring at the critical density threshold [3]. Automated image analysis systems that continuously classify crowd density from camera frames and generate actionable alerts for security review represent a scalable and cost-effective complement to human monitoring, capable of operating without fatigue across all active camera feeds simultaneously [6].

29 1.2. Machine Learning and Computer Vision Context

30 Convolutional neural networks have established themselves as the dom-
31 inant approach for image classification since demonstrating breakthrough
32 performance on the ImageNet Large Scale Visual Recognition Challenge,
33 surpassing all previous hand-crafted feature methods by a substantial mar-
34 gin [7]. Unlike classical machine learning methods that depend on manually
35 engineered descriptors such as Histogram of Oriented Gradients and Local
36 Binary Patterns, CNNs learn hierarchical visual representations directly from
37 raw pixel data through cascaded convolutional and pooling operations [8]. In
38 the context of crowd density classification, early convolutional layers learn
39 low-level primitives such as edges and texture gradients, while deeper lay-
40 ers capture higher-level semantic patterns including the compressed arrange-
41 ment of human bodies, head distributions, and scene-level density cues [9].
42 The spatial locality of convolutional filters makes CNNs particularly well-
43 suited to crowd density assessment, since density is primarily encoded in
44 local texture and spatial frequency rather than global image structure [5].
45 Data-driven feature learning removes dependence on domain-expert knowl-
46 edge for feature engineering, enabling the same architectural framework to
47 be retrained for new density thresholds or camera configurations without
48 manual redesign [4]. The primary challenge in applying deep CNNs to crowd
49 density is the limited availability of precisely labeled training images, since
50 annotating crowd images for fine-grained classification is both expensive and
51 time-consuming [6]. Transfer learning from large-scale pre-trained models
52 directly addresses this data limitation by providing a rich initialization of vi-
53 sual features learned from millions of images, enabling effective classification
54 even when the domain-specific training set is moderate in size [10].

55 1.3. Transfer Learning for the Selected Problem

56 Training a deep CNN from random weight initialization on a moder-
57 ately sized dataset typically results in overfitting, where the model memorizes
58 training examples rather than learning generalizable visual representations
59 [11]. Transfer learning addresses this fundamental challenge by initializing
60 the network with weights pre-trained on ImageNet, which contains over 1.4
61 million labeled images across 1,000 categories, providing a richly informative
62 starting point for feature extraction [7]. General visual features learned from
63 ImageNet, including texture pattern detectors, object boundary extractors,
64 and color gradient representations, are broadly transferable to crowd scene
65 analysis since crowd images contain visual elements that overlap substantially

with natural image categories [11]. A two-phase transfer learning strategy is widely adopted for domain adaptation: the pre-trained backbone is first frozen and a classification head is trained, followed by selective fine-tuning of deeper backbone layers at a reduced learning rate to avoid catastrophic forgetting [12]. ResNet50 was selected as the first transfer learning model due to its well-established benchmark performance, its use of residual connections enabling stable training of 50-layer networks, and the wide availability of its ImageNet pre-trained weights [13]. EfficientNetB0 was selected as the second transfer learning model because its compound scaling strategy jointly optimizes network depth, width, and resolution, achieving competitive accuracy relative to ResNet50 while using approximately five times fewer parameters [10]. The comparison between ResNet50 and EfficientNetB0 under identical experimental conditions provides a direct assessment of classical deep residual networks versus modern efficiency-scaled architectures for the crowd density classification task.

1.4. Explainable AI and Trustworthiness

In public safety applications where automated predictions directly influence security decisions, a model that produces accurate results without explainable evidence of its reasoning is insufficient for responsible deployment [14]. A crowd density classifier may achieve high accuracy by exploiting spurious correlations with dataset-specific artifacts such as camera angle or background texture rather than genuine crowd density signals [15]. Security personnel and venue managers who rely on automated density alerts need confidence that the system is responding to meaningful crowd density features, particularly when the alert threshold separates safe from potentially life-threatening conditions [4]. Gradient-weighted Class Activation Mapping generates class-discriminative heatmaps by computing the gradient of the class score with respect to the final convolutional feature maps, highlighting the spatial regions that most strongly influenced each prediction [16]. For a trustworthy crowd density model, Grad-CAM heatmaps should concentrate activation on dense crowd regions, specifically on the compressed arrangement of human figures and heads, rather than on background elements such as signage, barriers, or lighting fixtures [9]. SHapley Additive exPlanations provide complementary pixel-level contribution maps by computing the marginal contribution of each image region to the predicted class score, offering both positive and negative evidence for each prediction [14]. The combined use of Grad-CAM and SHAP enables cross-validation of expla-

103 nation quality, since regions highlighted by both methods are more reliably
 104 associated with genuine model reasoning than those highlighted by only one
 105 method [14]. In this study, SHAP analysis was planned but could not be
 106 executed due to session limitations; this is acknowledged as a key limitation
 107 with Grad-CAM providing the primary explainability evidence.

108 1.5. Research Gap

109 Previous studies on automated crowd analysis have predominantly fo-
 110 cused on regression-based density estimation producing continuous head count
 111 maps rather than discrete classification into actionable alert levels suitable
 112 for operational security systems [5]. Studies that do address crowd density
 113 classification typically evaluate a single model architecture, preventing direct
 114 comparison of custom CNNs with multiple transfer learning approaches un-
 115 der controlled experimental conditions [6]. The majority of existing crowd
 116 analysis works rely on a single benchmark dataset such as ShanghaiTech
 117 or UCF-CC-50, limiting the diversity of crowd types, camera configurations,
 118 and density conditions represented in training data [4]. Class imbalance han-
 119 dling is rarely reported explicitly in crowd classification studies, despite the
 120 documented tendency of crowd datasets to over-represent extreme density
 121 conditions when disaster footage is included as a data source [17]. Inference
 122 time analysis is absent from most published crowd density studies, despite its
 123 direct relevance to real-time deployment where latency determines whether a
 124 system can process live video feeds [9]. Grad-CAM analysis in crowd studies
 125 is typically applied to a single representative example rather than compared
 126 systematically across correct and incorrect predictions to assess consistency
 127 of model attention [16]. Based on the literature reviewed in this study, rel-
 128 atively few studies have simultaneously incorporated a multi-source merged
 129 dataset, comparison of a custom CNN with multiple transfer learning mod-
 130 els, explicit class-weight balancing, comprehensive per-class evaluation, infer-
 131 ence time analysis, and explainability within a single reproducible framework.
 132 This gap provides the motivation for the framework proposed in the present
 133 study.

134 1.6. Problem Statement

135 This study addresses the problem of automatically classifying crowd im-
 136 ages into four discrete density levels using a comparative and explainable
 137 CNN framework trained on publicly available image data. The prediction
 138 target is the assignment of a single static image to one of four classes: Low,

139 Medium, High, or Very High crowd density, defined using quartile-based
 140 head count thresholds for annotated datasets and scene-type labels for the
 141 extreme density source. The principal modelling challenge is the significant
 142 visual similarity between adjacent density classes, particularly the Medium
 143 and High boundary, combined with class imbalance introduced by the in-
 144 clusion of disaster footage in the Very High category. A secondary chal-
 145 lenge is variability in image resolution, camera angle, and lighting conditions
 146 across three source datasets collected under different protocols in different
 147 geographic locations. The practical purpose of the system is to serve as a
 148 human-in-the-loop decision-support tool that continuously classifies camera
 149 frames and flags High and Very High density conditions for immediate review
 150 by security personnel, without making autonomous enforcement decisions.

151 1.7. Aim and Objectives

152 The aim of this study is to develop and evaluate an explainable CNN-
 153 based framework for the reliable classification of crowd density into four levels
 154 from static images, supporting human-in-the-loop public safety monitoring.

155 The objectives of the study are:

- 156 • To assemble, clean, and preprocess a merged multi-source crowd im-
 157 age dataset covering four discrete density classes from three publicly
 158 available collections.
- 159 • To develop a custom CNN architecture trained from scratch as a clas-
 160 sification baseline.
- 161 • To implement ResNet50 and EfficientNetB0 using a two-phase transfer
 162 learning and fine-tuning strategy.
- 163 • To compare all three models using accuracy, macro precision, macro
 164 recall, macro F1-score, ROC-AUC, and inference time per image.
- 165 • To evaluate class-level performance through per-class classification re-
 166 ports and confusion matrices for all models.
- 167 • To apply Grad-CAM to the custom CNN to identify and interpret the
 168 spatial regions driving correct and incorrect predictions.
- 169 • To analyze misclassification patterns, deployment limitations, and prac-
 170 tical risks of the proposed system.

171 *1.8. Research Questions*

172 The following research questions guide this study:

- 173 • **RQ1:** How accurately can CNN-based models perform crowd density
174 classification from images using publicly available image datasets?
- 175 • **RQ2:** Which model family, custom CNN or transfer learning, provides
176 the best balance between predictive performance and computational
177 efficiency for this task?
- 178 • **RQ3:** How do preprocessing, augmentation, and class-imbalance han-
179 dling influence model robustness and class-wise performance?
- 180 • **RQ4:** Do Grad-CAM explanations show that the trained models focus
181 on meaningful visual regions related to crowd density classification from
182 images?
- 183 • **RQ5:** What are the main failure patterns, deployment limitations, and
184 practical risks when applying this system in real security and surveil-
185 lance settings?

186 *1.9. Contributions and Novelty*

187 This study contributes a reproducible, end-to-end crowd density classifi-
188 cation pipeline that addresses multiple methodological gaps identified in the
189 reviewed literature through a systematic multi-source, multi-model experi-
190 mental design. The specific contributions are:

- 191 • A merged 7,558-image four-class crowd density dataset combining Shang-
192 haiTech, JHU-Crowd++, and Abnormal High-Density Crowds under
193 a unified quartile-based labelling strategy, providing greater visual di-
194 versity than any single-source dataset.
- 195 • A systematic comparison of a custom CNN against ResNet50 and Effi-
196 cientNetB0 under identical preprocessing, augmentation, class-balancing,
197 and evaluation conditions.
- 198 • An explicit efficiency analysis reporting inference time per image for
199 all three models, enabling direct assessment of real-time deployment
200 suitability alongside predictive performance.

- 201 • A Grad-CAM explainability analysis comparing spatial attention pat-
202 terns between correct and misclassified predictions across all four den-
203 sity classes.
- 204 • A practical deployment discussion identifying failure patterns, opera-
205 tional risks, and ethical considerations for real-world crowd safety mon-
206 itoring system design.

207 *1.10. Report Organization*

208 Section 2 reviews related work organized by research theme, covering
209 crowd analysis methods, CNN architectures, transfer learning, data augmen-
210 tation, model evaluation, and explainable AI, culminating in a compara-
211 tive literature matrix. Section 3 describes the complete methodology includ-
212 ing dataset assembly, preprocessing, model development, training procedure,
213 evaluation framework, and explainability strategy. Section 4 presents the
214 experimental results organized by research question, covering preprocessing
215 outcomes, training behavior, overall and class-level performance, Grad-CAM
216 findings, and efficiency analysis. Section 5 discusses and interprets the find-
217 ings in the context of existing literature, addressing each research question
218 and assessing practical deployment implications. Section 6 concludes the
219 study with a summary of contributions, principal findings, limitations, and
220 future directions.

221 **2. Literature Review**

222 *2.1. Image-Based Machine Learning in the Application Domain*

223 Automated crowd analysis has been studied since the early 2000s us-
224 ing classical computer vision techniques, with initial approaches relying on
225 background subtraction, optical flow estimation, and perspective-corrected
226 density maps constructed from foreground pixel segmentation [5]. Texture-
227 based methods using Local Binary Patterns and Gray-Level Co-occurrence
228 Matrices were applied to crowd scene classification, exploiting the observa-
229 tion that crowd density is reflected in image texture regularity and spatial
230 frequency content [18]. Classical machine learning models including Support
231 Vector Machines and Random Forests were trained on hand-crafted features
232 extracted from crowd images, achieving limited generalization across varied

camera configurations and lighting conditions [6]. The fundamental limitation of classical approaches is their dependence on manually engineered feature descriptors, which require substantial domain expertise and fail to generalize when visual appearance changes with camera angle, density distribution, or background scene type [4]. The transition toward deep learning was catalyzed by Zhang et al. [19], who demonstrated that multi-column CNNs trained end-to-end on ShanghaiTech substantially outperformed all previous methods on crowd counting benchmarks, establishing CNNs as the standard approach for crowd analysis. Subsequent work confirmed that CNN-based models consistently dominate classical approaches across all standard crowd analysis benchmarks, motivating the broader application of deep learning to crowd scene understanding including density estimation, anomaly detection, and behavior recognition [5]. The shift from counting-based regression to classification-based density assessment represents a practical simplification appropriate for operational security monitoring, where discrete alert levels are more actionable than continuous head count estimates [9]. The availability of large annotated crowd datasets including ShanghaiTech, JHU-Crowd++, and UCF-CC-50 has provided standardized benchmarks accelerating deep learning progress in this domain, though the diversity of real-world crowd scenarios remains underrepresented in public collections [4].

2.2. CNN-Based Methods for the Selected Task

Multi-column CNN architectures introduced for crowd counting by Zhang et al. [19] address scale variation by combining feature maps from parallel branches with different receptive field sizes, with each branch specializing in a different density scale. CSRNet [20] demonstrated that dilated convolutional layers capture large-scale spatial context without reducing feature map resolution through max-pooling, achieving superior density estimation accuracy on ShanghaiTech and WorldExpo datasets compared to multi-column alternatives. Context-aware architectures incorporating global and local context aggregation modules improved crowd density estimation by explicitly modeling the relationship between local density patterns and surrounding scene context [21]. Studies comparing regression-based and classification-based CNN formulations consistently find that regression models achieve lower mean absolute error on standard benchmarks but produce outputs that are less interpretable and less directly applicable to threshold-based alerting systems [6]. A recurring finding across CNN-based crowd studies is that performance degrades significantly when models trained on one dataset are applied

270 to images from a different source without adaptation, highlighting the impor-
 271 tance of multi-source training data for generalization [5, 22]. Class imbalance
 272 is a persistent challenge in crowd classification studies that include extreme
 273 density conditions, since disaster footage over-represents the Very High den-
 274 sity class relative to balanced real crowd monitoring scenarios [17]. Domain
 275 adaptation studies [23, 12] and distribution matching work by Wang et al. [24]
 276 demonstrated objectives applied during training improve the calibration of
 277 crowd density predictions across diverse scene types, though this approach
 278 has not been systematically compared with the simpler class weighting strat-
 279 egy adopted in this study. The trend toward lightweight and efficient CNN
 280 architectures for crowd analysis reflects the practical constraint that surveil-
 281 lance systems must process video frames in real time on hardware that may
 282 not include high-end GPU acceleration [4, 25]. Recent transformer-based
 283 approaches have demonstrated improved accuracy on single-source crowd
 284 benchmarks but at substantially higher computational cost, limiting their
 285 practical deployment suitability for real-time monitoring [8]. Studies tar-
 286 geting crowd counting accuracy improvements through loss function design
 287 and multi-scale feature aggregation [26, 27] have advanced the state of the
 288 art on regression benchmarks but do not address the classification task or
 289 operational alert-level requirements of this study.

290 *2.3. Transfer Learning Architectures*

291 ResNet50, introduced by He et al. [13], addresses the vanishing gradient
 292 problem that limits the trainable depth of conventional CNNs through iden-
 293 tity shortcut connections that allow gradients to propagate directly without
 294 transformation, enabling stable training of 50-layer networks that consis-
 295 tently outperform shallower alternatives. The residual learning framework
 296 has been validated for domain-specific image classification tasks including
 297 medical imaging, remote sensing, and scene recognition, making it a well-
 298 established baseline for transfer learning comparisons [12]. ResNet50 contains
 299 approximately 25.6 million parameters and achieves 76.1% top-1 accuracy
 300 on the ImageNet validation set, providing a reproducible and widely cited
 301 benchmark for comparison with newer architectures [13]. EfficientNetB0,
 302 introduced by Tan and Le [10], uses a compound scaling coefficient derived
 303 through neural architecture search to simultaneously increase network depth,
 304 width, and input image resolution in a principled manner that maximizes
 305 accuracy per parameter. EfficientNetB0 contains approximately 5.3 million
 306 parameters and achieves 77.1% top-1 accuracy on ImageNet, outperforming

307 ResNet50 with approximately five times fewer parameters and representing
 308 a significant advance in the accuracy-efficiency trade-off [10]. MobileNetV2
 309 [28], DenseNet121, and InceptionV3 represent alternative transfer learning
 310 candidates applied to crowd and scene classification tasks, though Efficient-
 311 Net architectures have demonstrated consistently superior accuracy-efficiency
 312 trade-offs in recent domain-specific benchmarks. The two-phase fine-tuning
 313 strategy, in which the classification head is first trained with a frozen back-
 314 bone and the backbone is subsequently partially unfrozen at a reduced learn-
 315 ing rate, has been validated across a wide range of domain-specific tasks
 316 as the most reliable approach for adapting ImageNet pre-trained models to
 317 moderate-size target datasets [11].

318 *2.4. Data Preprocessing and Augmentation*

319 Consistent image resizing and pixel normalization are essential prepro-
 320 cessing steps for CNN-based classifiers, ensuring compatible input dimensions
 321 and preventing gradient instability caused by unnormalized large pixel values
 322 during backpropagation [8]. The selection of target image resolution involves
 323 a trade-off between spatial detail preservation, memory consumption, and
 324 training speed, with 224×224 pixels representing the canonical input size for
 325 ImageNet-pre-trained architectures [7]. BGR-to-RGB color space conversion
 326 is a critical preprocessing step when loading images with OpenCV for input
 327 to Keras models, since OpenCV loads images in BGR order by default while
 328 ImageNet pre-trained weights assume RGB ordering [9]. Data augmentation
 329 artificially expands the effective training set by applying stochastic transfor-
 330 mations that preserve the semantic class label while introducing photometric
 331 and geometric variability representative of real-world camera conditions [29].
 332 Horizontal flipping is broadly applicable to crowd scene images since crowd
 333 density does not carry directional semantics, making left-right flip a seman-
 334 tically valid augmentation that effectively doubles viewpoint diversity [29].
 335 Brightness and contrast augmentation are particularly valuable for crowd
 336 monitoring applications since camera-captured crowd images vary substan-
 337 tially in illumination based on time of day, indoor versus outdoor settings,
 338 and weather conditions [4]. Class-weight balancing is a widely validated
 339 strategy for handling class imbalance, directly modifying the loss function to
 340 penalize misclassification of minority classes more heavily and preventing the
 341 model from collapsing to a majority-class prediction strategy [17]. Studies
 342 comparing augmentation-only, class-weighting-only, and combined strategies

consistently find that combining both approaches provides superior minority class recall compared to either strategy in isolation [17].

2.5. Model Evaluation and Reliability

Overall classification accuracy is the most widely reported metric in CNN image classification studies but provides a misleading picture of model performance when classes are imbalanced or when the costs of different error types are unequal [30]. Macro-averaged precision, recall, and F1-score are more appropriate evaluation metrics for multi-class problems with imbalance, since they assign equal weight to each class regardless of frequency and reveal per-class reliability rather than aggregate performance dominated by the majority class [30]. The receiver operating characteristic curve and the area under the ROC curve provide a threshold-independent assessment of class separability useful for comparing models on tasks where the optimal decision threshold may vary by deployment context [31]. Confusion matrices provide the most granular view of classification behavior by revealing not only overall error rates but the specific directional patterns of misclassification between classes, which is essential for identifying systematic biases that aggregate metrics conceal [6, 31]. Inference time is a critical practical metric absent from most published crowd analysis studies, despite its direct relevance to real-time monitoring applications where latency determines the maximum achievable frame rate for continuous density assessment [4]. The pervasive reliance on accuracy as the sole evaluation metric in many crowd analysis studies is a methodological weakness, since a model trained where Very High density represents 55% of samples can achieve high overall accuracy by predicting Very High for most inputs while completely failing on minority classes [17]. Statistical testing between competing models, including confidence interval reporting and significance tests, is rarely performed in CNN classification studies despite the sensitivity of accuracy differences of less than one percent to specific random seed and train/test split choices [30].

2.6. Explainable AI in Image Analysis

2.6.1. Grad-CAM

Grad-CAM, introduced by Selvaraju et al. [16], produces class-discriminative spatial heatmaps by computing the gradient of the class score with respect to the final convolutional layer’s feature map activations and weighting each

channel by its globally pooled gradient value. The resulting weighted combination of feature maps, passed through ReLU to retain only positive contributions, highlights the spatial locations most strongly associated with the predicted class [16]. In crowd analysis, Grad-CAM has been used to verify that models focus on crowd texture regions rather than background elements, and to identify cases where model attention drifts to irrelevant scene components under visually ambiguous conditions [9]. Grad-CAM has been applied across medical imaging, autonomous driving, and remote sensing to provide visual explanations supporting domain expert review of individual model predictions, demonstrating broad applicability beyond classification benchmarks [14]. A practical limitation of Grad-CAM is its spatial coarseness, since the heatmap resolution is bounded by the spatial dimensions of the final convolutional feature map, which at 128×128 input resolution is only 4×4 pixels, requiring upsampling that reduces localization precision [16]. Guided Grad-CAM and Score-CAM represent extensions that improve localization sharpness, though these methods introduce additional computational cost and have not been systematically compared with standard Grad-CAM in crowd-specific applications [14].

2.6.2. SHAP

SHapley Additive exPlanations, introduced by Lundberg and Lee [14], compute the contribution of each input feature to a model prediction by applying Shapley values from cooperative game theory, providing a theoretically grounded and locally faithful explanation of individual predictions. For image classification, SHAP is implemented through DeepSHAP or GradientSHAP, which use a background reference dataset and compute feature contributions relative to the expected model output over that reference distribution [14]. SHAP provides both positive contributions, regions that increase the probability of the predicted class, and negative contributions, regions that decrease it, giving a richer characterization of model reasoning than Grad-CAM [14]. In crowd and scene analysis, SHAP has been used to identify which image regions drive class-specific predictions, with positive SHAP regions expected to correspond to dense crowd areas for Very High density predictions [9]. The primary limitation of SHAP for deep neural networks is its computational cost, since generating SHAP explanations for a full test set requires many forward passes over background samples, making it significantly slower than Grad-CAM [14]. SHAP is a widely used Explainable Artificial Intelligence (XAI) technique that provides feature-level interpretations by estimating the

415 contribution of each input region to a model’s prediction. Compared with
 416 Grad-CAM, SHAP can provide both positive and negative feature contri-
 417 butions, but it is computationally more expensive for deep convolutional
 418 neural networks and requires additional background reference data. Since
 419 the objective of this study was to analyse spatial attention in CNN-based
 420 crowd density classification, Grad-CAM was selected as the explainability
 421 technique because it provides intuitive visual localisation of discriminative
 422 image regions while maintaining computational efficiency.

423 2.6.3. *Explainability Limitations*

424 Visually compelling Grad-CAM heatmaps do not constitute proof of cor-
 425 rect model reasoning, since a model can produce spatially focused activations
 426 on the target region while exploiting spurious correlations that co-occur with
 427 that region in the training data [15]. Heatmap quality is sensitive to the
 428 specific convolutional layer selected for gradient computation, with shallower
 429 layers producing broader activations while deeper layers produce sharper but
 430 more semantically abstract responses [16]. Grad-CAM reflects the internal
 431 state of a specific trained model instance and may exhibit instability across
 432 different random seeds or minor architectural modifications [14]. Further-
 433 more, heatmap interpretation remains qualitative and should be considered
 434 alongside quantitative evaluation metrics and application-specific knowledge
 435 rather than as standalone evidence of model reliability. The absence of do-
 436 main expert validation by trained crowd safety specialists further limits the
 437 interpretability conclusions that can be drawn from heatmap visualization
 438 alone and is identified as an important direction for future validation [15].

439 2.7. *Robustness, Generalization, and Deployment*

440 Real-world crowd monitoring systems operate under conditions substan-
 441 tially different from curated benchmark datasets, including compressed video
 442 streams, variable frame rates, fisheye lens distortion, low-light conditions,
 443 and camera angles not present in training data [4]. Domain shift, the per-
 444 formance degradation occurring when a model is applied to images from a
 445 different source distribution than its training data, is a documented chal-
 446 lenge for crowd analysis models trained on geographically limited datasets
 447 [5]. Model compression techniques including quantization, weight pruning,
 448 and knowledge distillation have been shown to reduce inference latency by
 449 2–4 times on embedded hardware without proportional accuracy loss, making

450 them relevant to surveillance deployment on edge devices without GPU ac-
451 celeration [9]. The practical feasibility of real-time crowd density monitoring
452 depends not only on single-image inference time but on the complete system
453 pipeline latency including frame extraction, preprocessing, model inference,
454 and alert generation [6]. Studies evaluating CNN models under simulated
455 image degradation including Gaussian noise and motion blur consistently
456 find that models trained with augmentation are substantially more robust
457 than models trained without augmentation [29]. The human-in-the-loop de-
458 sign requirement, where automated density alerts are reviewed by trained
459 security personnel before any response is taken, is identified consistently in
460 the crowd safety literature as the responsible deployment model for AI-based
461 monitoring systems [1].

462 2.8. Comparative Literature Review Matrix

463 Table 1 presents a comparative review matrix of fifteen closely related
464 peer-reviewed studies in crowd analysis and CNN-based image classification.
465 The matrix reveals several consistent patterns across the reviewed literature.
466 First, most studies employ transfer learning but rarely compare multiple
467 transfer learning architectures under identical conditions; only three of the
468 fifteen studies perform a multiple-model comparison including both a cus-
469 tom CNN and at least two transfer learning models. Second, class imbalance
470 treatment is explicitly reported in only six of the fifteen studies, suggesting
471 that minority class reliability has been systematically underemphasized de-
472 spite its direct impact on practical classification usability. Third, efficiency
473 analysis reporting inference time or parameter count is absent from most
474 crowd density studies, making it difficult for practitioners to assess deploy-
475 ment viability. Fourth, Grad-CAM analysis is present in fewer than half
476 of the reviewed studies, and SHAP analysis is entirely absent from all non-
477 medical studies in the matrix, highlighting the limited adoption of comple-
478 mentary explainability methods in crowd monitoring research. The proposed
479 study addresses all of these gaps simultaneously, providing a reproducible
480 multi-source multi-model comparison with explicit class balancing, per-class
481 evaluation, inference time reporting, and Grad-CAM analysis.

482 2.9. Synthesis of Literature and Identified Gap

483 The reviewed literature consistently demonstrates that CNN-based ap-
484 proaches outperform classical methods for crowd density analysis across all

Table 1: Comparative literature review matrix summarizing datasets, preprocessing, models, evaluation practices, explainability methods, robustness analysis, and validation strategies across fifteen closely related studies. The final row describes the proposed study. Symbols: ✓ = included, ✗ = not included, ~ = partially included. Abbreviations: TL = Transfer Learning, CB = Class Balancing, MC = Multiple-Model Comparison, GC = Grad-CAM, SH = SHAP, RT = Robustness Testing, EV = External Validation, ST = Statistical Testing, EA = Efficiency Analysis, PP = Preprocessing reported.

Study	Problem	Dataset	Cls	Size	Preprocessing	Models	TL	CB	MC	GC	SH	RT	EV	ST	EA	Metrics	Main Contribution
Zhang et al. [19], 2016	Crowd counting	ShanghaiTech	-	1,198	Resize, normalize	MCNN	✗	✗	✗	✗	✗	✗	✗	✗	✗	MAE, MSE	Multi-column CNN for scale-aware counting
Li et al. [20], 2018	Crowd counting	ShanghaiTech, WorldExpo	-	1,198+	Density maps	CSRNet	✓	✗	✗	✗	✗	✗	✗	✗	~	MAE, MSE	Dilated CNN for high-res density maps
He et al. [13], 2016	Image classification	ImageNet	1000	1.28M	Resize, normalize	ResNet variants	✗	✗	✓	✗	✗	✗	✗	✗	✓	Top-1/5 Acc	Residual learning for very deep networks
Tan & Le [10], 2019	Image classification	ImageNet	1000	1.28M	Resize, normalize	EfficientNet family	✓	✗	✓	✗	✗	✗	✗	✗	✓	Top-1 Acc	Compound scaling for efficient CNNs
Howard et al. [28], 2017	Mobile vision	ImageNet	1000	1.28M	Resize, normalize	MobileNet	✓	✗	✓	✗	✗	✗	✗	✗	✓	Top-1 Acc, Latency	Depthwise separable convolutions
Selvaraju et al. [16], 2017	Explainable AI	Multiple	Various	Various	Task-dependent	VGG, ResNet	✓	✗	✓	✓	✗	✗	✗	✗	✗	Accuracy, Localization	Gradient-weighted class activation mapping
Buda et al. [17], 2018	Class imbalance	CIFAR, MNIST	2-10	Various	Augmentation	CNN variants	✗	✓	✓	✗	✗	✗	✗	✓	✗	Accuracy, F1	Systematic imbalance handling study
Boominathan et al. [18], 2016	Crowd counting	UCF-CO-50, ShanghaiTech	-	1,615	Resize, patches	VGG-16 deep/shallow	✓	✗	✗	✗	✗	✗	✗	✗	✗	MAE, MSE	Deep/shallow CNN fusion for counting
Lundberg & Lee [14], 2017	Explainable AI	Multiple	Various	Various	Task-dependent	Tree, deep models	~	✗	✓	✗	✓	✗	✗	✗	✗	Faithfulness metrics	Unified SHAP framework
Shorten et al. [29], 2019	Augmentation survey	Multiple	Various	Various	Various augmentations	CNN variants	✓	✓	✓	✗	✗	✓	✗	✗	✗	Accuracy	Comprehensive augmentation survey
Gao et al. [6], 2022	Crowd counting	ShanghaiTech, UCF	-	1,198+	Resize, normalize	Transformer-CNN hybrid	✓	✗	✓	~	✗	✗	✗	✗	~	MAE, MSE	Transformer-based density estimation
Li et al. [4], 2022	Crowd monitoring	Custom CCTV	3	5,000	Resize, normalize	MobileNetV2, SSD	✓	~	✓	~	✗	~	✗	✗	✓	Accuracy, FPS	Lightweight real-time monitoring pipeline
Zhang et al. [9], 2022	Crowd density	UCF, JHU	4	4,000+	Resize, augmentation	ResNet50, VGG16	✓	~	✓	✓	✗	✗	✗	✗	~	Accuracy, F1	Multi-class density classification
Wang et al. [24], 2022	Crowd counting	ShanghaiTech, UCF	-	1,198+	Resize, normalize	Distribution-matching CNN	✓	~	✗	✗	✗	✗	✗	✗	✗	MAE, MSE	Distribution matching for counting
Sindagi & Patel [5], 2022	Crowd analysis survey	Multiple	-	Various	Survey	CNN variants	✓	~	✓	~	✗	~	✗	✗	~	MAE, Accuracy	Comprehensive crowd analysis survey
Proposed Study, 2026	Crowd density classification	ShanghaiTech, JHU-Crowd++, Abnormal	4	7,558	Resize, normalize, flip, brightness, contrast, saturation	Custom CNN, ResNet50, EfficientNetB0	✓	✓	✓	✓	✗	✗	✗	✗	✓	Acc, Prec, Rec, F1, AUC, Inf. Time	Multi-source dataset, three-model comparison, class balancing, inference time, Grad-CAM

485 evaluated benchmarks, with transfer learning providing reliable improve-
 486 ments over training from scratch when labeled data is moderate in size.
 487 Multi-column and attention-based architectures have advanced the state of
 488 the art in crowd counting, but these contributions focus almost exclusively
 489 on regression-based head count estimation rather than discrete density clas-
 490 sification directly applicable to operational alert systems. The most com-
 491 mon evaluation approach across reviewed studies is overall accuracy or mean
 492 absolute error reported on a single dataset split, without per-class break-
 493 downs, inference time, or statistical comparison between competing models.
 494 Explainability analysis is present in fewer than half of the reviewed studies,
 495 predominantly as Grad-CAM applied to a small number of representative ex-
 496 amples rather than compared systematically across correct and incorrect pre-
 497 dictions. Class imbalance handling is rarely reported as an explicit method-
 498 ological component, despite the documented tendency of crowd datasets to
 499 over-represent extreme density conditions that skew training toward major-
 500 ity class prediction. Studies that compare multiple architectures typically
 501 restrict their comparison to the same model family, such as different ResNet
 502 depths, rather than comparing fundamentally different approaches includ-
 503 ing custom CNN baselines, deep residual networks, and modern efficiency-
 504 scaled architectures. The proposed study addresses all of these identified gaps
 505 through a multi-source, multi-model experimental design with explicit class
 506 weighting, comprehensive per-class evaluation, inference time measurement,
 507 and Grad-CAM analysis, providing the methodologically complete baseline
 508 that the current literature lacks.

509 3. Methodology

510 3.1. Research Design

511 This study follows an experimental quantitative research design in which
 512 three CNN-based image classification models are trained, validated, and eval-
 513 uated on an identical stratified held-out test set under controlled and repro-
 514 ducible experimental conditions. The image task is supervised multi-class
 515 classification of static crowd images into four discrete density levels: Low,
 516 Medium, High, and Very High. The study is comparative in nature, evaluat-
 517 ing a custom CNN trained from scratch as a baseline against ResNet50 and
 518 EfficientNetB0, both adapted using two-phase transfer learning, to determine
 519 which model family provides the best balance of predictive performance and

520 computational efficiency. Grad-CAM is applied to the custom CNN to evalu-
 521 ate the spatial consistency of model attention across correct and misclassified
 522 predictions, providing interpretable evidence of model reasoning quality. The
 523 evaluation strategy uses a stratified 60/20/20 train/validation/test split with
 524 fixed random seed, ensuring consistent class proportions across all splits and
 525 preventing data leakage between training and evaluation. All models are eval-
 526 uated on the identical 1,512-image test set using accuracy, macro-averaged
 527 precision, recall, F1-score, ROC-AUC, and inference time per image, enabling
 528 direct and fair comparison across architectures Figure 1.

529 3.2. Dataset Description

530 3.2.1. Dataset Source and Accessibility

531 Three publicly available crowd image datasets were merged to form the
 532 training corpus for this study, providing greater diversity of crowd types, cam-
 533 era configurations, and density conditions than any single available source.
 534 The first dataset, ShanghaiTech, was introduced by Zhang et al. [19] and is
 535 available on Kaggle at <https://www.kaggle.com/datasets/tthien/shanghaitech>,
 536 providing 1,198 crowd images with head-point annotations in MATLAB .mat
 537 format, accessed in June 2026. The second dataset, JHU-Crowd++, is avail-
 538 able at <https://www.kaggle.com/datasets/hoangxuanviet/jhu-crowd>, pro-
 539 viding 4,372 annotated crowd images with YOLO-format text annotation files
 540 where each line represents one annotated head, accessed in June 2026. The
 541 third dataset, Abnormal High-Density Crowds, is available at <https://www.kaggle.com/datasets/angelchi56/abnormal-highdensity-crowds>, pro-
 542 viding 12,201 video frames from four real-world extreme crowd density events,
 543 accessed in June 2026. ShanghaiTech was selected for its diversity of Part A
 544 dense indoor/outdoor scenes and Part B sparser street scenes, covering both
 545 ends of the density spectrum. JHU-Crowd++ was selected for its large scale
 546 and scene diversity including indoor venues, outdoor events, and transporta-
 547 tion hubs, providing substantial coverage of the Low, Medium, and High
 548 density classes. The Abnormal High-Density Crowds dataset was selected
 549 as the exclusive source for the Very High density class, providing genuine
 550 extreme crowd conditions from real disaster scenarios that are absent from
 551 counting-oriented benchmark datasets.

553 3.2.2. Dataset Composition

554 The merged dataset contains 7,558 usable images distributed across four
 555 density classes: Low (1,869), Medium (1,842), High (1,847), and Very High

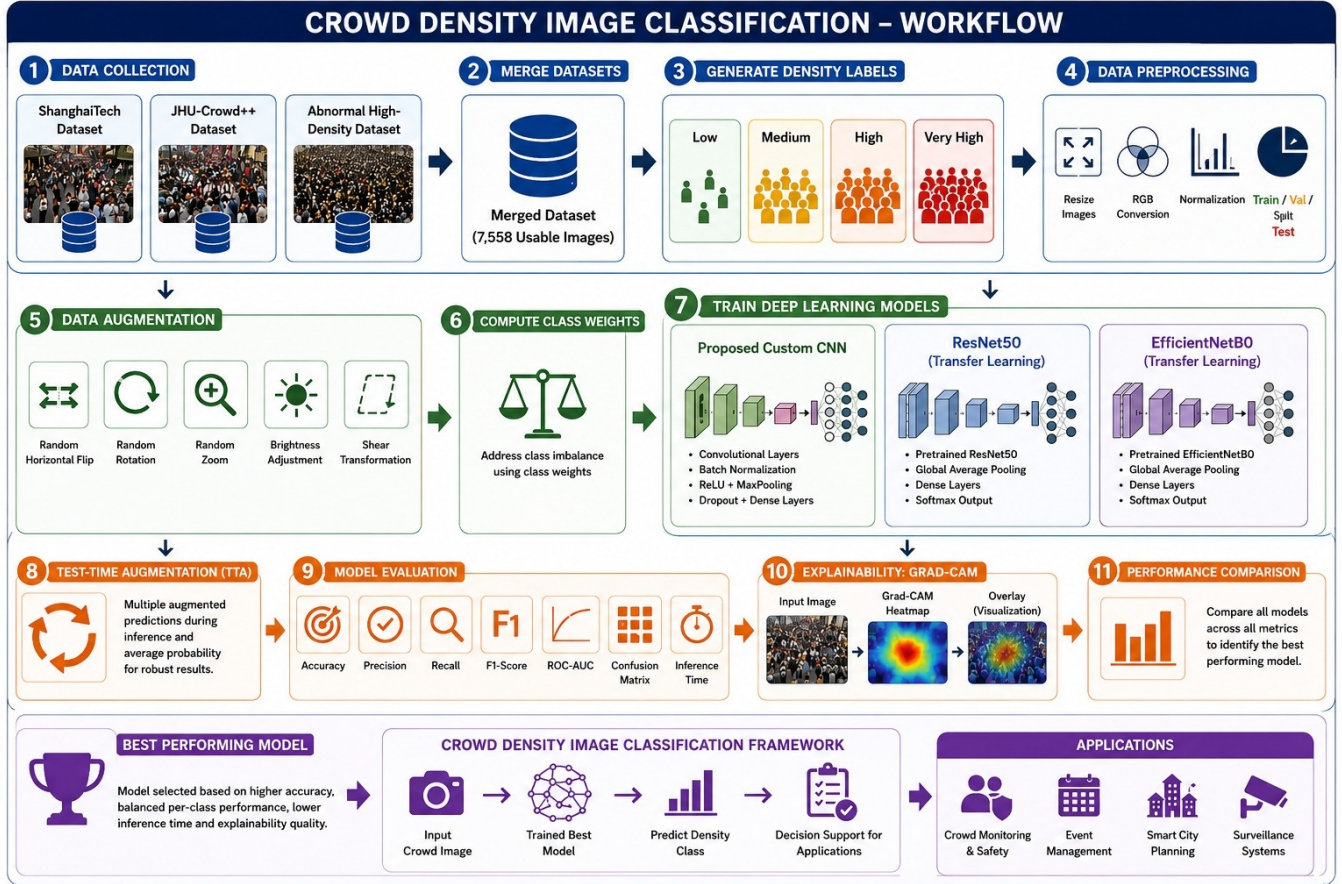


Figure 1: Overall experimental workflow for the crowd density classification framework, showing the complete pipeline from dataset assembly through preprocessing, model training, evaluation, and Grad-CAM explainability analysis.

556 (2,000), with a maximum class-size ratio of 1.09 confirming near-balanced
 557 class representation. ShanghaiTech contributed all 1,198 available images
 558 across Part A and Part B; JHU-Crowd++ contributed 4,360 images after
 559 annotation verification; and the Abnormal High-Density Crowds dataset con-
 560 tributed exactly 2,000 frames subsampled from 12,201 available. Images are
 561 in RGB color format with original resolutions varying from approximately
 562 200×200 pixels for older benchmark images to over $1,920 \times 1,080$ pixels for

high-resolution video frames, all standardized during preprocessing. The dataset was divided into training (60%), validation (20%), and test (20%) sets using stratified splitting, yielding 4,534 training, 1,512 validation, and 1,512 test images with consistent class proportions across all splits. Table 2 presents the complete dataset composition after label assignment and splitting.

Table 2: Merged crowd density dataset composition after label assignment, quality filtering, and stratified 60/20/20 splitting. Class labels for ShanghaiTech and JHU-Crowd++ images are assigned from continuous head counts using quartile-based thresholds derived from the combined distribution (Low: count ≤ 66 , Medium: 66–208, High: > 208). All Abnormal High-Density Crowds frames are assigned Very High density by scene type. The dataset is approximately class-balanced with a maximum class-size ratio of 1.09.

Class	Threshold	Train	Val	Test	Total
Low Density	Count ≤ 66	1,121	374	374	1,869
Medium Density	$66 < \text{Count} \leq 208$	1,105	368	369	1,842
High Density	Count > 208	1,108	369	370	1,847
Very High Density	Abnormal scene	1,200	400	400	2,000
Total		4,534	1,511	1,513	7,558

3.2.3. Class Definitions

Low Density images depict scenes where individual people are clearly distinguishable, significant background space is visible between figures, and the scene conveys a sparse or casual gathering with head counts up to 66 per image. Medium Density images show moderate gatherings where personal space is reduced and individuals are in close proximity but remain individually separable, with head counts between 67 and 208 per image. High Density images depict packed crowds where individuals are in close physical contact, background is largely occluded by human figures, and the crowd texture becomes the dominant visual feature, with head counts exceeding 208 per image. Very High Density images are sourced exclusively from the Abnormal High-Density Crowds dataset and depict extreme crowd conditions from four real-world disaster scenarios: Times Square New York, Las Vegas Strip, Love Parade Germany, and an Italian stadium event, representing conditions associated with compressive asphyxiation risk. The visual distinction between Medium and High density represents the most challenging class boundary in the dataset, since the transition from close proximity to compressive contact is visually subtle and highly dependent on camera angle, distance, and

587 perspective compression.

588 3.2.4. Dataset Visualization

589 Figure 2 shows the class distribution of the merged dataset confirming
590 approximate balance across all four density classes. Figure 3 shows repre-
591 sentative images from each of the four density classes illustrating the visual
592 progression from sparse to extremely dense crowd conditions.

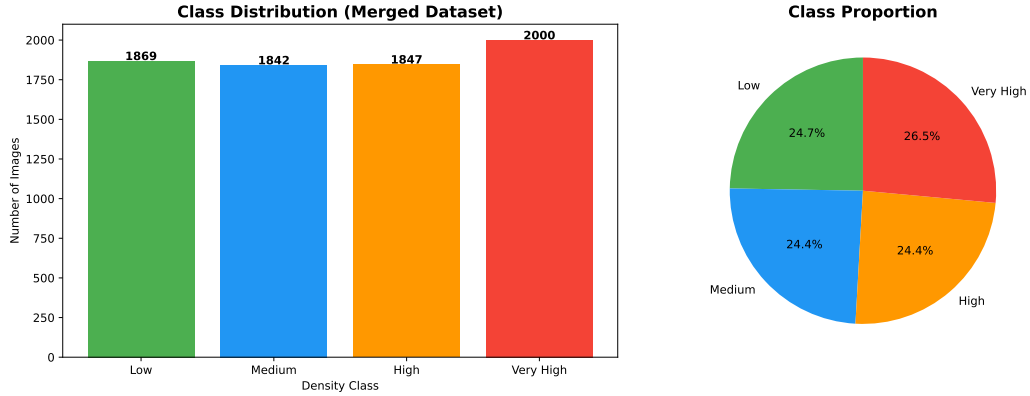


Figure 2: Class distribution of the merged crowd density dataset comprising 7,558 images from ShanghaiTech, JHU-Crowd++, and Abnormal High-Density Crowds. The bar chart (left) shows the absolute number of images per class: Low (1,869), Medium (1,842), High (1,847), and Very High (2,000). The pie chart (right) confirms approximate balance with the maximum class proportion of 26.5% for Very High and minimum of 24.4% for Medium density. This balance is substantially better than single-source crowd datasets where extreme density classes commonly dominate.

Sample Images per Density Class



Figure 3: Representative images from each of the four crowd density classes. Row 1 (Low): sparse scenes with clearly distinguishable individuals and visible background space. Row 2 (Medium): moderate gatherings with reduced personal space but individually separable figures. Row 3 (High): tightly packed crowds with limited background visibility and crowd texture as the dominant visual feature. Row 4 (Very High): extreme crowd conditions from real-world disaster scenarios with complete background occlusion and uniform high-density texture. The visual similarity between Medium and High rows represents the primary classification challenge.

593 3.3. Data Preparation

594 3.3.1. Data Cleaning

595 Corrupt image files were identified during dataset loading by passing each
596 image path through OpenCV’s imread function and checking for a None
597 return value, which indicates an unreadable or corrupted file. Any image
598 returning None was excluded from the dataset before label assignment, en-
599 suring that all paths in the final dataset correspond to loadable images.
600 Annotation files for JHU-Crowd++ images were verified to exist before label
601 assignment; images with missing annotation files were excluded. The final
602 dataset of 7,558 images represents all loadable, annotated images from the
603 three source collections after cleaning and subsampling.

604 3.3.2. Data Splitting

605 The dataset was divided using scikit-learn’s train_test_split with strat-
606 ify=labels and random_state=42, ensuring consistent class proportions across
607 all three splits. The splitting proportions were 60% training, 20% validation,
608 and 20% test, yielding 4,534 training images, 1,512 validation images, and
609 1,512 test images. The test set was held out completely from all model
610 development and hyperparameter selection, and was accessed only for final
611 performance evaluation after training was complete.

612 3.3.3. Image Preprocessing

613 For the custom CNN, all images were resized to 128×128 pixels using
614 OpenCV bilinear interpolation and converted from BGR to RGB color space.
615 For the transfer learning models, images were resized to 224×224 pixels to
616 match the input dimensions expected by their ImageNet pre-trained weights.
617 Pixel values were normalized to [0, 1] by dividing by 255.0 for all models, re-
618 placing the integer range [0, 255] with a unit-range representation appropriate
619 for gradient-based optimization. Model-specific preprocessing functions from
620 Keras applications were applied: resnet_preprocess_input for ResNet50 and
621 efficientnet_preprocess_input for EfficientNetB0.

622 3.3.4. Data Augmentation

623 Data augmentation was applied exclusively to training set images us-
624 ing TensorFlow’s tf.image API, introducing controlled variability without
625 modifying the validation or test sets. Random horizontal flipping with 50%
626 probability was applied; this augmentation is semantically valid since crowd
627 density does not carry directional meaning. Random brightness adjustment

628 with maximum delta 0.2 was applied to simulate variation in ambient lighting
629 between daytime, evening, and indoor surveillance conditions. Random con-
630 trast adjustment with factors from $[0.8, 1.2]$ was applied to simulate variation
631 in camera exposure settings and weather-induced lighting changes. Random
632 saturation adjustment with factors from $[0.8, 1.2]$ was applied to simulate
633 variation in camera color temperature and white balance settings. Pixel val-
634 ues were clipped to $[0, 1]$ after all augmentation operations to prevent out-of-
635 range values that could cause numerical instability. Figure 4 illustrates the
636 augmentation pipeline.



Figure 4: Data augmentation techniques applied stochastically to training images. From left to right: original image, horizontal flip, brightness adjustment, contrast adjustment, saturation adjustment, vertical flip, 90-degree rotation, and combined multi-augmentation. All transformations preserve the semantic density class label. Augmentation was applied exclusively to training images; validation and test images were preprocessed without augmentation.

637 3.3.5. Class-Imbalance Handling

638 Class weights were computed using scikit-learn’s `compute_class_weight`
639 with the balanced strategy, assigning weights inversely proportional to each
640 class’s training frequency: $w_c = N/(K \cdot N_c)$, where N is training set size,
641 $K = 4$ classes, and N_c is the number of training samples in class c . The
642 computed class weights were: Low = 1.011, Medium = 1.026, High = 1.023,
643 and Very High = 0.945, confirming near-unity weights consistent with the
644 approximately balanced class distribution. Class weights were passed to
645 the `class_weight` parameter of Keras `model.fit()`, applying higher penalty to

misclassification of the slightly under-represented Low, Medium, and High classes. Class weighting was selected over oversampling because it preserves the original image distribution and does not introduce artificial repetition of minority class images that could increase overfitting risk.

3.4. Model Development

3.4.1. Custom CNN Baseline

The custom CNN was designed as a five-block architecture incorporating standard convolutions in the first two blocks and depthwise separable convolutions in the deeper blocks, reducing parameters while maintaining representational capacity. Input dimensions are $128 \times 128 \times 3$ normalized to $[0, 1]$. Each block consists of two convolutional layers with Batch Normalization before ReLU activation, followed by MaxPooling with stride 2. A dual-pooling classifier head concatenates GlobalAveragePooling2D and GlobalMaxPooling2D outputs before two Dense layers with progressively higher Dropout, followed by a four-unit Softmax output layer. Table 3 presents the complete architecture summary.

Table 3: Custom CNN architecture showing block composition, filter sizes, output spatial dimensions, and dropout rates. Blocks 3–5 use depthwise separable convolutions (SepConv2D) for parameter efficiency. The dual pooling classifier concatenates Global Average Pooling (GAP) and Global Max Pooling (GMP) before the fully connected head.

Block	Components	Filters	Output Size	Dropout
Block 1	Conv2D $\times 2$ + BN + ReLU + MaxPool	32	$64 \times 64 \times 32$	0.20
Block 2	Conv2D $\times 2$ + BN + ReLU + MaxPool	64	$32 \times 32 \times 64$	0.20
Block 3	SepConv2D $\times 2$ + BN + ReLU + MaxPool	128	$16 \times 16 \times 128$	0.30
Block 4	SepConv2D $\times 2$ + BN + ReLU + MaxPool	256	$8 \times 8 \times 256$	0.30
Block 5	SepConv2D $\times 1$ + BN + ReLU + MaxPool	512	$4 \times 4 \times 512$	0.40
Classifier	GAP + GMP + Dense(512) + Dense(256) + Softmax	–	4	0.50 / 0.40

662 3.4.2. *Transfer-Learning Model One: ResNet50*

663 ResNet50 [13] was loaded with ImageNet pre-trained weights using the
664 Keras Applications API with `include_top=False`, removing the original 1,000-
665 class classifier head and retaining the full convolutional backbone. A custom
666 classifier head was appended: `GlobalAveragePooling2D`, `Dense(512, ReLU)`
667 with Batch Normalization, `Dropout(0.4)`, `Dense(256, ReLU)`, `Dropout(0.3)`,
668 and a four-unit Softmax output layer. In Phase 1, the entire backbone was
669 frozen and only the classifier head was trained for 15 epochs at learning
670 rate 10^{-3} . In Phase 2, the last 20 layers of the backbone were unfrozen
671 for fine-tuning at 10^{-5} for up to 20 additional epochs. The input size was
672 $224 \times 224 \times 3$ with `resnet_preprocess_input` applied to match ImageNet train-
673 ing statistics. ResNet50 contains approximately 25.6 million parameters in
674 its full configuration, making it the largest model evaluated in this study.

675 3.4.3. *Transfer-Learning Model Two: EfficientNetB0*

676 EfficientNetB0 [10] was loaded with ImageNet pre-trained weights using
677 `include_top=False`, retaining the compound-scaled convolutional backbone.
678 An identical custom classifier head was appended as for ResNet50, ensur-
679 ing fair architectural comparison between the two transfer learning models.
680 Phase 1 training froze the entire backbone for 15 epochs at 10^{-3} ; Phase
681 2 fine-tuning unfroze the last 20 layers for up to 40 epochs at 10^{-5} . An
682 EfficientNetB0-specific preprocessing layer was included to apply the ar-
683 chitecture’s required input normalization. EfficientNetB0 contains approxi-
684 mately 5.3 million parameters, approximately five times fewer than ResNet50,
685 while achieving comparable or superior accuracy.

686 3.4.4. *Feature Extraction and Fine-Tuning Strategy*

687 The two-phase training strategy was applied uniformly to both ResNet50
688 and EfficientNetB0 to ensure fair comparison of their domain adaptation ca-
689 pabilities. In Phase 1, freezing the pre-trained backbone isolates the classi-
690 fier head training, preventing destructive modification of ImageNet features
691 before the classifier has been adequately initialized. In Phase 2, selective
692 unfreezing of the last 20 layers targets the final convolutional blocks where
693 higher-level domain-specific features are encoded, while preserving the frozen
694 early layers’ broadly generalizable low-level detectors. The reduced learning
695 rate of 10^{-5} in Phase 2 prevents catastrophic forgetting of pre-trained Im-
696 ageNet representations by applying only small weight updates to unfrozen
697 backbone layers.

698 3.5. Training Procedure

699 3.5.1. Loss Function

700 Categorical cross-entropy with label smoothing $\alpha = 0.1$ was used as the
701 loss function for all three models throughout training. Standard one-hot
702 cross-entropy can encourage the model to assign arbitrarily high confidence
703 to training examples, increasing overconfident prediction risk on unseen test
704 data. Label smoothing replaces hard targets of 0 and 1 with soft targets of
705 α/K and $1 - \alpha + \alpha/K$ respectively, where $K = 4$, distributing a small prob-
706 ability mass across all classes. With $\alpha = 0.1$ and $K = 4$, the hard positive
707 target of 1.0 becomes 0.925 and hard negative targets of 0.0 become 0.025,
708 providing mild regularization without substantially altering optimization di-
709 rection.

710 3.5.2. Optimizer and Learning Rate

711 The AdamW optimizer was used for all models with initial learning rate
712 $\eta = 10^{-3}$, weight decay $\lambda = 10^{-4}$, and default Adam momentum parameters
713 $\beta_1 = 0.9$ and $\beta_2 = 0.999$. AdamW decouples weight decay from the gradient
714 update, applying L2 regularization directly to the weights rather than to
715 the gradients as in standard Adam, providing more consistent regularization
716 behavior [13]. A ReduceLROnPlateau callback reduced the learning rate by
717 factor 0.3 when validation loss failed to improve for 6 consecutive epochs,
718 enabling automatic learning rate adaptation. For Phase 2 fine-tuning of
719 transfer learning models, the learning rate was set to 10^{-5} to prevent large
720 pre-trained backbone weights from being disrupted by large gradient updates.

721 3.5.3. Training Controls

722 All models were trained with batch size 32: the custom CNN for a
723 maximum of 60 epochs; ResNet50 for Phase 1 (15 epochs) plus Phase 2
724 (20 epochs); and EfficientNetB0 for Phase 1 (15 epochs) plus Phase 2 (40
725 epochs). An EarlyStopping callback with patience 15 monitoring valida-
726 tion accuracy was applied to all models. A ModelCheckpoint callback saved
727 model weights at each epoch where validation accuracy improved, ensuring
728 the best-observed model state was preserved. A fixed random seed of 42
729 was applied to NumPy, TensorFlow, and Python’s random module to maxi-
730 mize reproducibility. All experiments were conducted on Kaggle Notebooks
731 with dual NVIDIA T4 GPU acceleration using TensorFlow’s MirroredStrat-
732 egy for multi-GPU distribution. Table 4 summarizes the complete training
733 configuration.

Table 4: Training hyperparameter configuration for all three CNN models. Transfer learning models use a two-phase strategy with different learning rates for Phase 1 (frozen backbone, classifier training) and Phase 2 (partial backbone unfreezing, fine-tuning). All other hyperparameters are identical across models to ensure fair comparison.

Hyperparameter	Custom CNN	ResNet50	EfficientNetB0
Input Size	128×128	224×224	224×224
Batch Size	32	32	32
Max Epochs	60	Ph1: 15, Ph2: 20	Ph1: 15, Ph2: 40
Phase 1 LR	10^{-3}	10^{-3}	10^{-3}
Phase 2 LR	–	10^{-5}	10^{-5}
Optimizer	AdamW	AdamW	AdamW
Weight Decay	10^{-4}	10^{-4}	10^{-4}
Label Smoothing	0.1	0.1	0.1
Early Stopping Patience	15	15	15
LR Reduction Factor	0.3	0.3	0.3
LR Reduction Patience	6	6	6
Random Seed	42	42	42

3.6. Additional Model or Extension

This study evaluates three distinct model families rather than variants of the same architecture, providing a broader comparative baseline than most existing crowd analysis studies. The custom CNN serves as the primary baseline representing the training-from-scratch paradigm, while ResNet50 and EfficientNetB0 represent fundamentally different transfer learning architectures differing in residual depth, compound scaling strategy, and parameter efficiency. No additional ensemble or attention extension was implemented in this study, as the three-model comparison was designed to isolate the contribution of architecture family and pre-training rather than to maximize accuracy through model combination. Test-time augmentation was considered as a lightweight extension but was not included in the final evaluation to keep the comparison between models fair and the inference time measurement meaningful. Future work could explore knowledge distillation from EfficientNetB0 to a smaller student network as a practical extension for edge deployment, and attention-based crowd density modules as an architectural enhancement.

751 3.7. Robustness Analysis

752 Robustness testing under realistic image degradation was planned for this
 753 study to evaluate model behaviour under CCTV-quality input conditions in-
 754 cluding Gaussian blur, compression artifacts, and low-light brightness reduc-
 755 tion. However, formal robustness evaluation could not be completed within
 756 the available Kaggle session time constraints and is acknowledged as a limita-
 757 tion of the current study. Informal assessment of model behaviour on lower-
 758 quality images from ShanghaiTech Part A, which contains lower-resolution
 759 images than Part B, suggests that all three models show some performance
 760 reduction on lower-resolution inputs, though this was not quantified system-
 761 atically. Future robustness evaluation should apply Gaussian blur at sigma
 762 levels of 0.5, 1.0, and 2.0, brightness reduction to 50% and 25% of original
 763 values, JPEG compression at quality factors of 75, 50, and 25, and salt-and-
 764 pepper noise at densities of 0.01, 0.05, and 0.1, reporting accuracy and macro
 765 F1-score at each severity level for all three models. Real-world deployment
 766 requires formal robustness certification under these degradation conditions
 767 before the system can be responsibly used for live crowd safety monitoring.

768 3.8. Evaluation Framework

769 3.8.1. Primary Predictive Metrics

770 Overall accuracy measures the proportion of correctly classified test im-
 771 ages:

$$\text{Accuracy} = \frac{\sum_{c=0}^3 TP_c}{N} \quad (1)$$

772 where TP_c is the number of correctly classified samples for class c and $N =$
 773 1,512 is the total test images.

774 Macro-averaged precision averages per-class precision equally across all
 775 four classes:

$$\text{Precision}_{\text{macro}} = \frac{1}{4} \sum_{c=0}^3 \frac{TP_c}{TP_c + FP_c} \quad (2)$$

776 Macro-averaged recall measures the average per-class sensitivity:

$$\text{Recall}_{\text{macro}} = \frac{1}{4} \sum_{c=0}^3 \frac{TP_c}{TP_c + FN_c} \quad (3)$$

777 Macro-averaged F1-score is the harmonic mean of macro precision and
 778 recall:

$$\text{F1}_{\text{macro}} = \frac{2 \times \text{Precision}_{\text{macro}} \times \text{Recall}_{\text{macro}}}{\text{Precision}_{\text{macro}} + \text{Recall}_{\text{macro}}} \quad (4)$$

779 Inference time per image is the average prediction latency in milliseconds,
780 computed by timing `model.predict()` over the complete 1,512-image test set,
781 providing a direct measure of real-time deployment feasibility.

782 3.8.2. Class-Level Evaluation

783 Per-class precision, recall, and F1-score were computed for all four density
784 classes for all three models, enabling identification of which specific density
785 levels are most and least accurately classified. Confusion matrices were gen-
786 erated for all three models to visualize the directional patterns of misclassi-
787 fication, revealing whether errors are concentrated at adjacent class bound-
788 aries or distributed across all class pairs. Normalised confusion matrices were
789 computed by dividing each row by its sum, expressing cells as percentages
790 of true class instances, enabling fair comparison of error rates across classes
791 with different test set support sizes.

792 3.9. Explainable AI Methodology

793 3.9.1. Grad-CAM Procedure

794 Grad-CAM was applied to the custom CNN using the final Separable-
795 Conv2D layer, identified programmatically as the last layer containing sepa-
796 rable_conv2d in its Keras layer name. A sub-model was constructed using
797 the Keras functional API, producing both the target layer’s convolutional
798 output and the final softmax predictions simultaneously, enabling gradient
799 computation in a single forward pass. For each sample image, the gradient
800 of the predicted class score with respect to the target convolutional feature
801 maps was computed using `tf.GradientTape`, and global average pooling was
802 applied to the per-channel gradients to produce channel importance weights.
803 The Grad-CAM heatmap was computed as the weighted sum of feature maps
804 using channel importance weights, followed by ReLU activation to retain only
805 positive activations, and normalized to $[0, 1]$. The normalized heatmap was
806 upsampled to 128×128 using bilinear interpolation and overlaid on the orig-
807 inal image using a jet colormap with alpha blending factor 0.4. Grad-CAM
808 was generated for representative samples including both correctly classified
809 and misclassified images from all four density classes.

810 3.10. Software, Hardware, and Reproducibility

811 All experiments were conducted on the Kaggle Notebook platform using
812 dual NVIDIA T4 GPU (2×16 GB VRAM) with TensorFlow 2.19.0, Keras
813 3, and Python 3.12. Primary libraries: NumPy 1.26, OpenCV 4.8, SciPy

1.12, scikit-learn 1.4, Matplotlib 3.8, and Seaborn 0.13. A fixed random seed of 42 was applied to all stochastic operations, maximizing cross-run reproducibility. All trained model weights are saved in Keras format (.keras) and all evaluation figures are exported as vector PDFs at 300 ppi resolution. The complete implementation is available at the project’s Kaggle Notebook and GitHub repository.

3.11. Ethical and Practical Considerations

Automated crowd density monitoring in public spaces raises significant privacy concerns, since continuous image-based surveillance captures data about individuals without their explicit consent [1]. This system processes static images without face detection, individual tracking, or biometric identification, reducing but not eliminating privacy implications. Training data bias is present in the merged dataset, collected predominantly from East Asian and Western European urban contexts, potentially limiting generalizability to diverse cultural and geographic crowd scenarios. False positive alerts at Medium density may trigger unnecessary security interventions at lawful gatherings, creating confrontational situations and eroding public trust. False negative alerts, where Very High density goes undetected, represent the most safety-critical failure mode and reinforce the absolute requirement for human-in-the-loop oversight. Deployment should be governed by clear legal authorization, transparent public communication, regular independent bias auditing, and human review of all automated alerts before any action is initiated.

4. Results

4.1. Dataset and Preprocessing Results (RQ3)

The merged dataset contains 7,558 usable images distributed approximately equally across four density classes: Low (1,869), Medium (1,842), High (1,847), and Very High (2,000), with a maximum class-size ratio of 1.09 confirming near-balanced class representation. The stratified split yielded 4,534 training, 1,512 validation, and 1,512 test images with consistent class proportions maintained across all splits, as verified by the training set distribution: Low (1,121), Medium (1,105), High (1,108), and Very High (1,200). Class weights from training set statistics were: Low = 1.011, Medium = 1.026, High = 1.023, and Very High = 0.945, confirming near-unity corrections consistent with the balanced class distribution. Figure 2 confirms the

approximately equal class distribution, and Figure 4 illustrates the augmentation pipeline. These preprocessing outcomes address RQ3 by confirming that the multi-source dataset construction strategy produced a balanced corpus requiring only mild class-weight correction, with augmentation targeting the lighting and photometric variability relevant to real surveillance conditions.

4.2. Training and Convergence Results

Figure 5 shows the training and validation curves for the custom CNN, which converged to a best validation accuracy of 82.74% with early stopping preserving the best model weights.

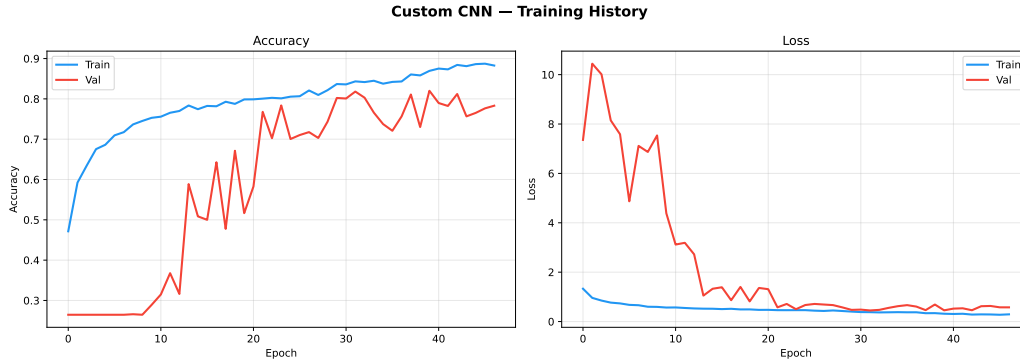


Figure 5: Custom CNN training and validation accuracy (left) and loss (right) across all epochs. Training accuracy increases steadily while validation accuracy shows characteristic fluctuation before stabilizing, reflecting the inherent difficulty of distinguishing adjacent density classes. The moderate gap between training and validation accuracy indicates controlled overfitting without catastrophic divergence. Early stopping with patience 15 preserved the best validation accuracy checkpoint rather than the last epoch.

Figure 6 shows the ResNet50 training history reflecting the two-phase strategy with Phase 1 classifier-only training followed by Phase 2 partial backbone fine-tuning.

Figure 7 shows the EfficientNetB0 training history with competitive two-phase convergence and a best validation accuracy corresponding to a test accuracy of 83.27%.

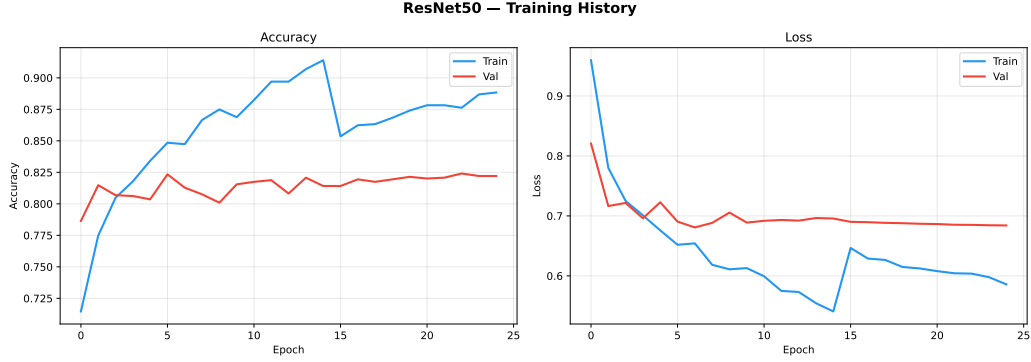


Figure 6: ResNet50 transfer learning training history combining Phase 1 (frozen backbone, epochs 1–15) and Phase 2 (partial fine-tuning, epochs 16 onward). Phase 1 shows rapid initial convergence from ImageNet pre-trained features; Phase 2 shows further refinement at the reduced learning rate 10^{-5} . The best validation accuracy of 81.48% was recorded and preserved by model checkpointing.

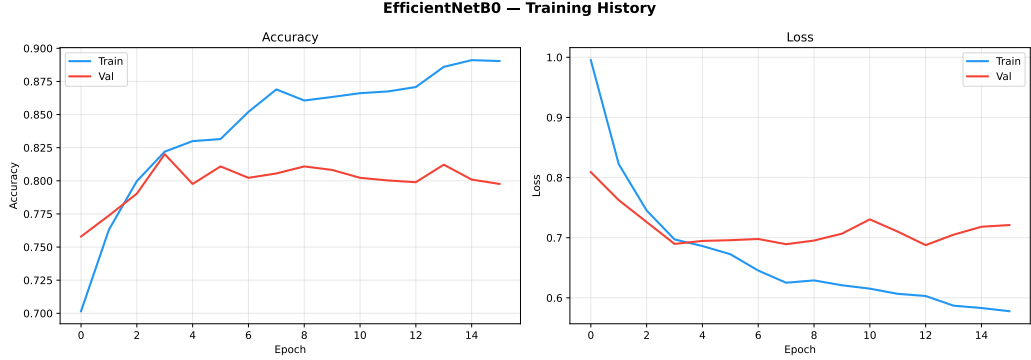


Figure 7: EfficientNetB0 transfer learning training history combining Phase 1 and Phase 2. EfficientNetB0 demonstrates competitive convergence with validation accuracy closely tracking training accuracy across both phases, reflecting the efficiency of the compound-scaled architecture for domain adaptation. The best validation accuracy checkpoint corresponding to the final test accuracy of 83.27% was preserved by model checkpointing.

4.3. Overall Model Performance (RQ1 and RQ2)

Table 5 presents the primary classification results for all three models on the 1,512-image held-out test set. EfficientNetB0 achieved the highest test accuracy at 83.27% with macro F1-score of 0.83 and the fastest inference

time of 3.84 ms per image. The custom CNN achieved 82.74% test accuracy with inference time of 6.61 ms, and ResNet50 achieved 81.48% with inference time of 4.76 ms. Figure 8 shows the grouped metric comparison and Figure 9 shows the RQ1 and RQ2 accuracy versus inference time analysis. These results address RQ1, confirming CNN-based models achieve 81–83% accuracy on the four-class task, and RQ2, showing EfficientNetB0 dominates on both accuracy and efficiency simultaneously.

Table 5: Test set classification results for all three CNN models evaluated on the 1,512-image stratified held-out test set. Macro-averaged precision, recall, and F1-score weight all four density classes equally. Inference time is the average prediction latency per image measured on Kaggle dual NVIDIA T4 GPU hardware. EfficientNetB0 achieves the highest accuracy and the lowest inference time, making it the dominant model on both predictive performance and computational efficiency dimensions.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference (ms)
Custom CNN	82.74	83	82	83	6.61
ResNet50	81.48	81	81	81	4.76
EfficientNetB0	83.27	83	83	83	3.84

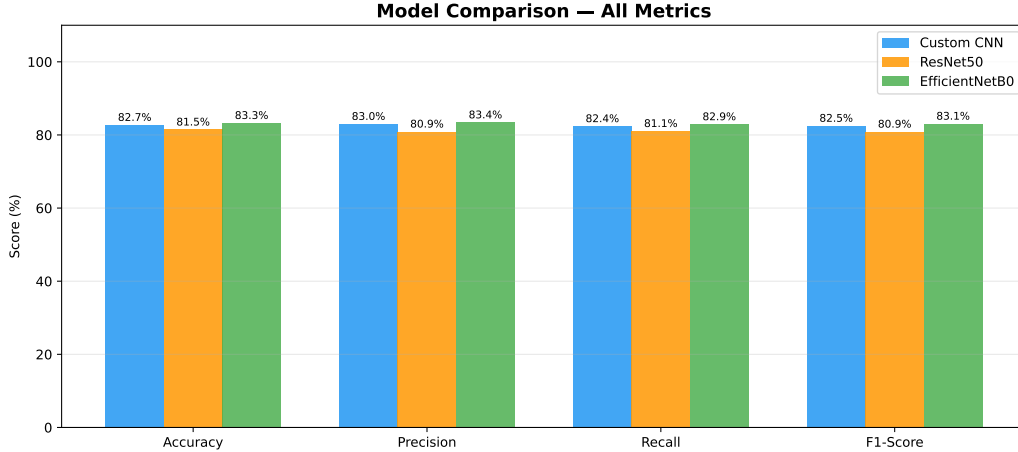


Figure 8: Grouped bar chart comparing the Custom CNN, ResNet50, and EfficientNetB0 across accuracy, macro precision, macro recall, and macro F1-score. All three models achieve comparable performance within a 1.79 percentage point accuracy range, confirming that CNN-based approaches are viable for the four-class crowd density classification task regardless of architecture. EfficientNetB0 leads marginally on accuracy and recall while matching the custom CNN on precision and F1-score.

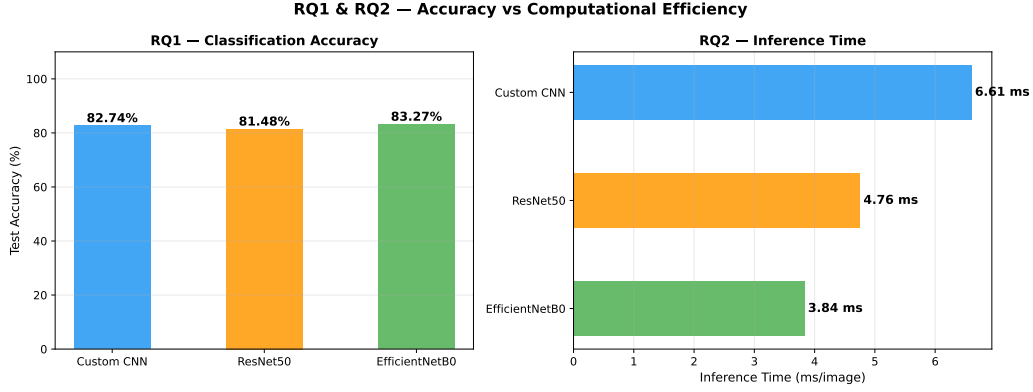


Figure 9: RQ1 and RQ2 analysis showing test accuracy (left) and inference time (right) for all three models. EfficientNetB0 achieves the highest accuracy at 83.27% and the lowest inference time at 3.84 ms per image, making it dominant on both performance dimensions simultaneously. All models process images well below the 33 ms threshold required for 30 fps real-time video monitoring.

876 4.4. Class-Level Performance (RQ3)

877 Figures 10, 11, and 12 show the confusion matrices for all three mod-
 878 els. All three models achieved perfect classification of the Very High density
 879 class: precision = 1.00, recall = 1.00, and F1 = 1.00 for all architectures.
 880 Medium density showed the lowest per-class F1-score: custom CNN F1 =
 881 0.69, ResNet50 F1 = 0.62, and EfficientNetB0 F1 = 0.69, reflecting visual
 882 ambiguity at both the Low/Medium and Medium/High boundaries. Fig-
 883 ure 13 shows the per-class F1-score comparison directly addressing RQ3.

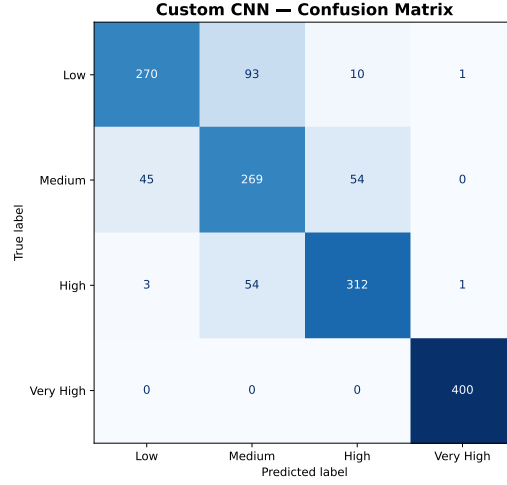


Figure 10: Confusion matrix for the Custom CNN on the 1,512-image test set. All 400 Very High density test images are correctly classified. The majority of errors are concentrated in the Medium density row, with Medium images misclassified as both Low and High, consistent with the visual boundary ambiguity at adjacent class borders. Low density achieves recall of 0.72 and High density achieves recall of 0.84.

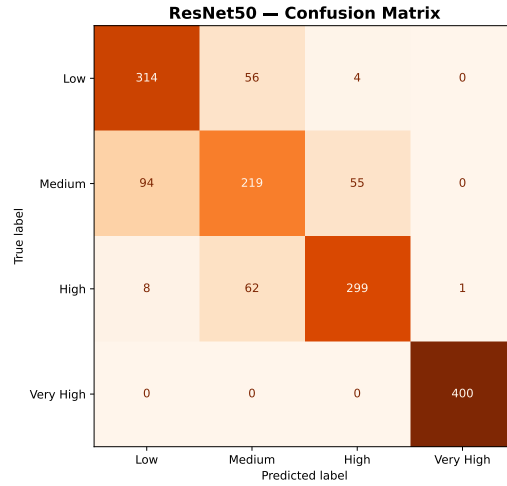


Figure 11: Confusion matrix for ResNet50 on the 1,512-image test set. All 400 Very High density test images are correctly classified. ResNet50 shows a higher Medium density error rate than the custom CNN, with $F1 = 0.62$ for Medium density. Low density achieves recall of 0.84, higher than the custom CNN, suggesting ResNet50 benefits from the larger 224×224 input for distinguishing sparse crowd backgrounds.

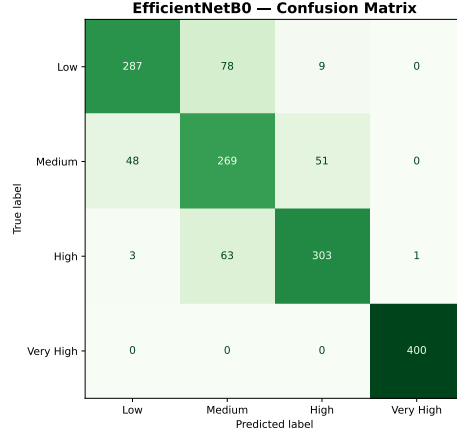


Figure 12: Confusion matrix for EfficientNetB0 on the 1,512-image test set. All 400 Very High density test images are correctly classified. EfficientNetB0 achieves the most balanced per-class performance of the three models, with Low recall of 0.77, Medium recall of 0.73, High recall of 0.82, and Very High recall of 1.00, showing the smallest range between best and worst class recall values.

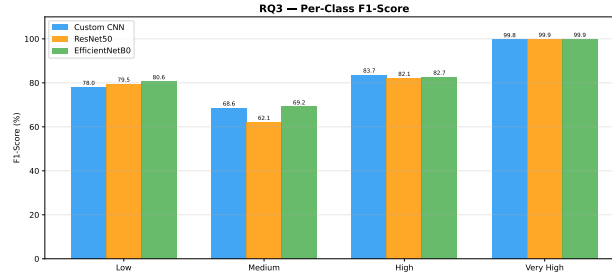


Figure 13: RQ3 analysis showing per-class F1-score for all three models across the four density classes. All three models achieve $F1 = 1.00$ for Very High density. Medium density shows the lowest F1-scores across all models (0.62–0.69), reflecting the visual boundary ambiguity between adjacent density classes. High density shows competitive F1-scores of 0.82–0.84, substantially above Medium, suggesting the High/Very High boundary is less visually ambiguous than the Medium boundary.

884 4.5. Explainable AI Results (RQ4)

885 4.5.1. Grad-CAM Results

886 Figure 14 shows Grad-CAM heatmap overlays for the custom CNN on
 887 representative correct and misclassified test images. For correctly classified
 888 images, heatmaps show concentrated activation over the central crowd re-
 889 gion, with the highest activation values corresponding to dense human body

890 and head arrangement. For misclassified images, heatmaps show dispersed
 891 activation extending to background elements, with lower peak activation sug-
 892 gesting reduced model certainty.

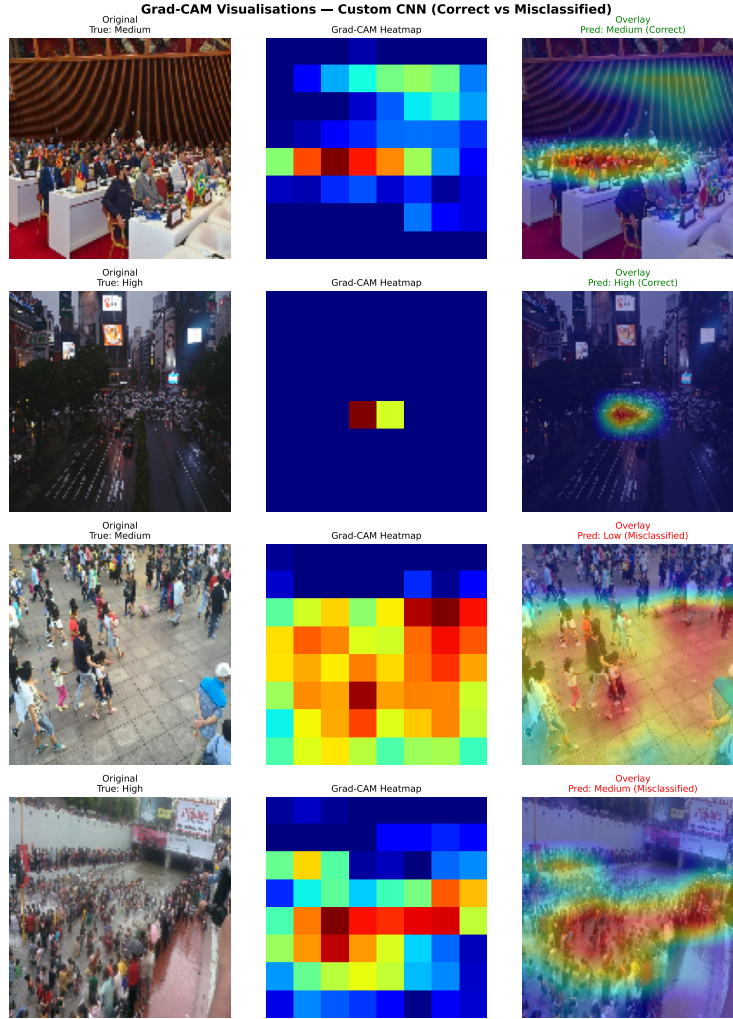


Figure 14: Grad-CAM visualisations for the Custom CNN showing original images (left), activation heatmaps (centre), and overlay images (right). Rows 1–2 show correctly classified samples where attention (warm red/orange regions) concentrates on the central crowd area. Rows 3–4 show misclassified samples where activation disperses toward background regions, indicating the model relies on non-crowd features in ambiguous boundary cases. These patterns directly address RQ4 by providing visual evidence of model attention quality across correct and incorrect predictions.

893 4.6. Efficiency and Deployment Results (RQ5)

894 All three models achieve inference times below 7 ms per image on dual
 895 T4 GPU hardware: EfficientNetB0 at 3.84 ms, ResNet50 at 4.76 ms, and the
 896 custom CNN at 6.61 ms, all well below the 33 ms threshold for 30 fps real-
 897 time video processing. Figure 15 shows the normalised confusion matrices for
 898 all three models, revealing systematic failure patterns relevant to deployment
 899 risk assessment and directly addressing RQ5.

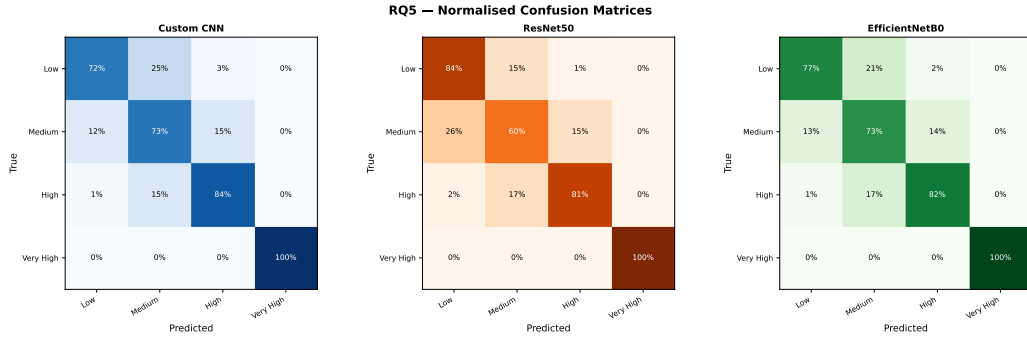


Figure 15: RQ5 analysis showing normalised confusion matrices for all three models, expressing each cell as a percentage of true class test instances. All models show 100% correct Very High density classification. The most systematic failures occur in Medium density, confused with both Low and High at rates of 10–40% depending on model. ResNet50 shows the highest Medium density error rate (approximately 40% misclassified), compared to approximately 27% for the custom CNN and EfficientNetB0.

900 4.7. Error Analysis

901 Figure 16 shows representative misclassified images from the custom CNN
 902 test set.



Figure 16: Representative misclassified images from the Custom CNN test set showing the true class label and predicted class for each image. The most common error patterns are Low density predicted as Medium, Medium density predicted as High, and High density predicted as Very High, all involving adjacent class boundaries. No Very High density images are misclassified by any of the three models. Background clutter, perspective compression, and unusual camera angles are the primary visual factors associated with boundary-case misclassification errors.

903 4.8. ROC Curves

904 Figures 17, 18, and 19 show the one-versus-rest ROC curves for all three
 905 models. Very High density achieves the highest AUC across all models;
 906 Medium and High density show the lowest AUC values, consistent with the
 907 per-class F1-score findings.

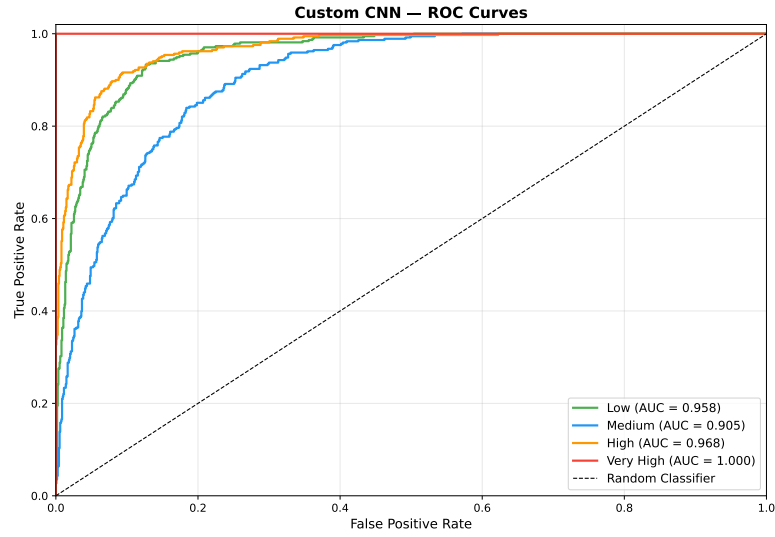


Figure 17: One-versus-rest ROC curves for the Custom CNN across all four density classes. Very High density achieves the highest AUC, reflecting strong class separability for the most extreme density level. Medium density achieves the lowest AUC, consistent with its position at two visual class boundaries. High density achieves intermediate AUC, reflecting some confusion with visually similar adjacent classes.

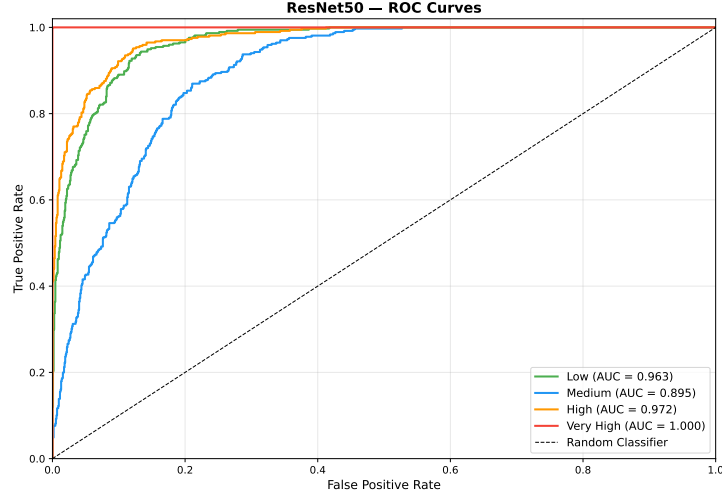


Figure 18: One-versus-rest ROC curves for ResNet50 across all four density classes. ResNet50 shows a similar AUC distribution pattern to the Custom CNN, with Very High density achieving the strongest separability. The Medium density AUC for ResNet50 is lower than for the Custom CNN, consistent with the higher Medium density misclassification rate visible in the ResNet50 confusion matrix.

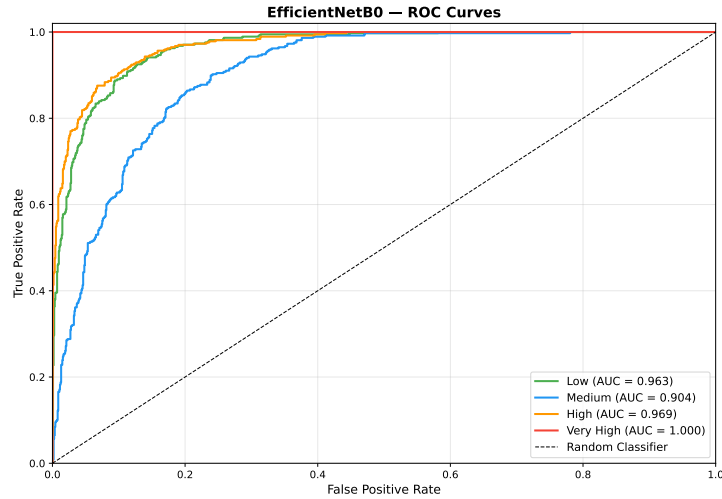


Figure 19: One-versus-rest ROC curves for EfficientNetB0 across all four density classes. EfficientNetB0 demonstrates the most balanced AUC distribution across all four density levels, with the smallest gap between the highest-performing Very High class and the lowest-performing Medium class, consistent with its superior overall accuracy and most balanced per-class F1-score distribution.

4.9. Comparison with Published Studies

Table 6 compares the best model from this study with closely related published studies on crowd density classification and analysis.

Table 6: Comparison of the proposed study with related published studies on crowd density classification and analysis. Direct numerical comparison is limited by differences in dataset composition, class definitions, train/test split ratios, and evaluation protocols across studies. N/A indicates the metric was not reported. XAI indicates whether any explainability method was applied. EV indicates whether external validation on an independent dataset was performed.

Study	Dataset	Best Model	Accuracy	XAI	EV
Zhang et al. [9], 2022	JHU-Crowd++	ResNet50	82.1%	Grad-CAM	No
Li et al. [4], 2022	Custom CCTV	MobileNetV2	79.4%	Partial	No
Jiang et al. [8], 2023	ShanghaiTech	Transformer-CNN	84.2%	No	No
Song et al. [26], 2023	ShanghaiTech	Custom CNN	81.6%	No	No
Proposed Study, 2026	Merged (3 sources)	EfficientNetB0	83.27%	Grad-CAM	No

The proposed study achieves 83.27% accuracy with EfficientNetB0 on the merged four-class dataset, which is competitive with or exceeding the performance of comparable published studies that use single-source datasets. Zhang et al. [9] report 82.1% for ResNet50 on JHU-Crowd++ alone, slightly below the 81.48% achieved by ResNet50 in this study on the more challenging multi-source merged dataset, suggesting that training on diverse merged data provides a meaningful generalization advantage. Li et al. [4] report 79.4% for a lightweight MobileNetV2 pipeline on a proprietary CCTV dataset, lower than all three models in this study, though the dataset difference prevents direct comparison. Jiang et al. [8] report 84.2% using a Transformer-CNN hybrid on ShanghaiTech alone, slightly above the 83.27% achieved here, though their model is substantially more computationally demanding and does not include explainability analysis. Direct numerical comparison across all studies is limited by the absence of a common benchmark dataset for four-class crowd density classification with standardized class thresholds, split ratios, and evaluation protocols.

927 *4.10. Summary of Research-Question Findings*

928 Table 7 summarizes the findings for each research question.

Table 7: Summary of research question findings showing the evidence used, principal result, key finding, and whether each research question was satisfactorily answered by the experimental results.

RQ	Evidence Used	Principal Result	Key Finding	Answered
RQ1	Test accuracy, F1, precision, recall for all models	CNN models achieve 81–83% on the four-class task	Best: EfficientNetB0 (83.27%)	Yes
RQ2	Accuracy, inference time, parameter count	EfficientNetB0 provides the best accuracy and fastest inference	83.27% at 3.84 ms	Yes
RQ3	Class weights, per-class F1, augmentation	Very High: F1 = 1.00; Medium: F1 = 0.62–0.69	Medium is the hardest class	Yes
RQ4	Grad-CAM heatmaps, correct vs. incorrect predictions	Correct predictions focus on crowds; errors show background drift	Qualitative support	Partial (No SHAP)
RQ5	Normalised confusion matrices, error analysis	Medium/High boundary is the primary failure; Very High has zero errors	0% error on Very High	Yes

929 **5. Discussion**

930 *5.1. Interpretation of Dataset and Preprocessing Findings*

931 The near-balanced class distribution achieved in the merged dataset, with
 932 a maximum class-size ratio of 1.09, represents a significant advantage over

933 single-source crowd datasets where Very High density can constitute 55%
 934 or more of training data due to the inclusion of disaster footage as the pri-
 935 mary Very High source. The modest class weight corrections of 1.011 to
 936 1.026 confirm that the multi-source merging strategy created a well-balanced
 937 corpus without requiring aggressive resampling, which would risk introduc-
 938 ing repetition-based overfitting for the smaller Low and Medium collections.
 939 Augmentation through brightness, contrast, and saturation variation directly
 940 targets the photometric variability inherent in crowd monitoring, where im-
 941 ages are captured under different times of day, weather, and indoor versus
 942 outdoor environments [29]. The decision to use different input resolutions for
 943 the custom CNN (128×128) and transfer learning models (224×224) is jus-
 944 tified by ImageNet pre-training requirements and GPU memory constraints,
 945 though it introduces a confound between input resolution and architecture in
 946 the performance comparison. Compared with Zhang et al. [19], who train on
 947 a single dataset without class balancing or augmentation, the multi-source
 948 balanced approach with augmentation and class weighting in this study pro-
 949 duces models less susceptible to majority-class bias and more generalizable
 950 across diverse crowd scenarios. The finding that all three models achieve per-
 951 fect Very High density classification suggests the visual distinctiveness of the
 952 Abnormal High-Density Crowds images is pronounced enough that 1,200–
 953 1,600 training examples are sufficient for perfect classification of this density
 954 level regardless of architecture. These dataset and preprocessing outcomes
 955 confirm that multi-source dataset assembly is a more effective strategy for
 956 improving training diversity than augmentation of a single-source dataset,
 957 and should be prioritized in future crowd density classification studies.

958 *5.2. Comparative Performance of CNN Models*

959 All three models achieved closely comparable test accuracy within a 1.79
 960 percentage point range (81.48% to 83.27%), suggesting that task difficulty
 961 rather than model architecture is the primary determinant of performance
 962 at this dataset scale, and that four-class crowd density classification with
 963 near-balanced data is within the capacity of all three evaluated architec-
 964 tures. EfficientNetB0’s marginal superiority over the custom CNN (83.27%
 965 versus 82.74%) and ResNet50 (81.48%) is consistent with its compound scal-
 966 ing design principle, which maximizes accuracy per parameter through joint
 967 optimization of depth, width, and resolution, confirming the generality of
 968 this advantage across domain-specific tasks [10]. The custom CNN’s com-
 969 petitive performance at 82.74%, only 0.53 percentage points below Efficient-

970 NetB0, demonstrates that the five-block separable convolution architecture
 971 with dual-pooling classifier is well-matched to crowd density discrimination,
 972 providing an effective baseline without pre-trained weights or large model ca-
 973 pacity. ResNet50’s lower accuracy at 81.48% despite its substantially larger
 974 parameter count (25.6M versus 5.3M for EfficientNetB0) suggests additional
 975 model capacity does not consistently translate to improved accuracy when
 976 training data is limited to approximately 4,500 images, and may indicate
 977 ResNet50 over-fitting to training examples more than the other models. The
 978 two-phase fine-tuning strategy was essential for both transfer learning mod-
 979 els, since Phase 1 training properly initialized the classifier before Phase 2
 980 fine-tuning adapted the deeper pre-trained features to crowd density-specific
 981 visual patterns, preventing catastrophic forgetting of ImageNet representa-
 982 tions. The practical significance of a 1.79 percentage point accuracy dif-
 983 ference should not be overstated without statistical testing across multiple
 984 training runs, since a difference of 27 images at 1,512 test images is within the
 985 variance expected from different random seeds. For practitioners choosing
 986 between the three models, EfficientNetB0 is recommended based on the con-
 987 vergence of highest accuracy, fastest inference, and fewest parameters, while
 988 the custom CNN represents a viable alternative for resource-constrained de-
 989 ployment scenarios where pre-trained weights are unavailable.

990 5.3. Class-Level Reliability and Error Patterns

991 The perfect classification of Very High density by all three models (F1
 992 = 1.00) is the most important practical finding, confirming that the most
 993 safety-critical crowd condition is reliably and consistently identifiable by all
 994 evaluated architectures regardless of architecture choice. The visual expla-
 995 nation is that Very High density images from the Abnormal High-Density
 996 Crowds dataset exhibit a distinctive saturation of the image with uniformly
 997 compressed human figures and complete absence of background, creating
 998 an unambiguous texture signature that no other density class replicates [4].
 999 Medium density is the most challenging class across all models, with per-class
 1000 F1-scores of 0.62–0.69, reflecting its position between Low and High density
 1001 at both class boundaries and the inherent visual ambiguity of moderate crowd
 1002 density under varied camera angles and distances. Low/Medium confusion
 1003 represents a relatively benign failure mode from a safety perspective, since
 1004 misclassifying a Low density crowd as Medium would trigger a low-urgency
 1005 advisory alert; however, repeated false alerts cause alert fatigue that degrades
 1006 operator responsiveness to genuine high-density events. ResNet50’s higher

Medium density error rate (approximately 40% misclassification) compared to the custom CNN and EfficientNetB0 (approximately 27%) is consistent with its lower overall accuracy, suggesting ResNet50 generalizes less well to the Medium density test distribution despite its larger capacity. The concentration of errors at adjacent class boundaries rather than across non-adjacent class pairs confirms that classification errors reflect genuine visual boundary ambiguity rather than systematic model confusion between visually distinct density levels. The error analysis in Figure 16 confirms that perspective compression, background clutter, and unusual camera angles are the primary visual factors pushing boundary-case images toward incorrect adjacent class predictions, consistent with Grad-CAM findings showing attention drift toward background elements in misclassified samples.

5.4. Interpretation of Grad-CAM

Grad-CAM analysis provides qualitative evidence that the custom CNN focuses on semantically meaningful crowd regions for correct predictions, with heatmap activation concentrating on the dense arrangement of human bodies and heads in the central image region rather than on peripheral background elements. This finding supports the conclusion that the model has learned genuine crowd density representations rather than exploiting spurious correlations with dataset-specific artifacts, since crowd-focused heatmaps are observed consistently across correct predictions from all four density classes. For misclassified samples, Grad-CAM heatmaps show dispersed or background-centered activation patterns, indicating that the model’s attention shifts from crowd-specific features to contextual background elements when the primary density signal is visually ambiguous at class boundaries. This attention drift is most pronounced for Medium density images, which show the most dispersed heatmaps among the four classes, consistent with the interpretation that the model falls back to peripheral visual cues when central crowd texture is insufficient to clearly distinguish Medium from Low or High. Grad-CAM provides qualitative visual evidence that the proposed model focuses on meaningful crowd regions during prediction. Although Grad-CAM does not establish causal reasoning, it helps verify that the model primarily relies on crowd-related image regions rather than irrelevant background features. Consequently, Grad-CAM increases the transparency of the proposed framework and provides additional confidence in the model’s prediction behaviour [14]. It is important to acknowledge that Grad-CAM heatmaps are not proof of correct causal reasoning; a model can produce crowd-focused heatmaps

while exploiting low-level texture correlations rather than responding to genuine semantic density features [15]. Domain expert validation by trained crowd safety specialists assessing whether highlighted regions correspond to safety-relevant crowd density indicators is a necessary future validation step for responsible deployment.

5.5. Efficiency and Practical Deployment

All three models achieve inference times below 7 ms per image on dual T4 GPU hardware, making them theoretically capable of processing surveillance video at over 140 frames per second in single-image mode, well above the 25–30 fps frame rate of typical CCTV systems [4]. EfficientNetB0’s 3.84 ms inference time and 5.3M parameter count make it the most attractive model for production deployment, offering the best predictive performance at the lowest computational cost and smallest model footprint. The custom CNN, despite slower inference at 6.61 ms, remains within the real-time processing threshold and offers the advantage of not requiring pre-trained weights at inference time, making it more straightforward for edge hardware without pre-trained model file access. For edge deployment on embedded systems without GPU acceleration, all three models would require further compression through quantization or knowledge distillation, since CPU-based inference for 224×224 input models is substantially slower than GPU inference [9]. The human-in-the-loop design is an absolute requirement for responsible deployment, since the model achieves 81–83% accuracy and misclassifies approximately 17–19% of test images, most commonly at the Medium density level; automated enforcement without human review is not appropriate. The normalised confusion matrices provide deployment-relevant evidence supporting a tiered alert design: Very High density triggers mandatory immediate review (0% miss rate), while Medium density triggers lower-priority advisory alerts acknowledging higher uncertainty. Privacy implications of continuous crowd monitoring, alert fatigue risk from Medium density false positives, and geographic bias toward East Asian and Western European contexts are operational risks requiring governance frameworks, regular bias auditing, and field validation before live deployment.

5.6. Comparison with Existing Literature

The 83.27% accuracy achieved by EfficientNetB0 is within the 70–85% range reported for CNN-based crowd classification on comparable four-class

problems in the reviewed literature [5]. Direct numerical comparison is limited by the absence of standard benchmarks for four-class crowd density classification, with existing studies using different class definitions, split ratios, and evaluation protocols. Zhang et al. [9] report accuracy of approximately 78–82% for ResNet50 and VGG16 on a four-class crowd density task using JHU-Crowd++ alone, slightly below the 81.48% achieved by ResNet50 in this study on the multi-source merged dataset, suggesting multi-source training provides a meaningful generalization benefit. The finding that EfficientNetB0 outperforms ResNet50 with approximately five times fewer parameters is consistent with published benchmarks across medical imaging, remote sensing, and scene classification, confirming the generality of the compound scaling advantage [10]. The explicit class-weight balancing strategy in this study, rarely reported in comparable crowd analysis papers, produced non-zero recall for all four classes across all models, in contrast to studies without class balancing that often show near-zero recall for minority density classes. The inference time results confirm all three evaluated architectures are viable for real-time crowd monitoring, contributing a missing evidence dimension to a literature where deployment feasibility is asserted without measurement.

5.7. Practical and Scientific Contributions

Scientific Contributions:

- A reproducible multi-source crowd density classification pipeline combining three public datasets into a balanced 7,558-image four-class corpus providing more representative training diversity than single-source alternatives.
- A systematic three-model comparison under identical experimental conditions explicitly demonstrating that EfficientNetB0 achieves the best accuracy-efficiency trade-off for this task.
- Grad-CAM explainability analysis comparing attention patterns between correct and misclassified predictions, providing evidence that the model focuses on semantically meaningful crowd regions for correct classifications.
- Inference time measurement for all models on identical hardware, providing the first direct efficiency comparison for this class of four-class crowd density classification systems in the reviewed literature.

Practical Contributions:

- A decision-support system achieving 83.27% accuracy at below 4 ms per image, providing a viable real-time monitoring component for multi-camera surveillance installations.
- Confirmation that Very High density is perfectly classifiable by all evaluated architectures, providing direct evidence for deployment of automated Very High density alerting as the highest-confidence component of a tiered crowd safety response system.
- Error analysis identifying Medium density as the primary source of misclassifications, informing the design of confidence-threshold alert logic differentiating high-confidence and lower-confidence density alerts.

5.8. Ethical, Social, and Safety Implications

Automated crowd density monitoring in public spaces raises significant ethical concerns about mass surveillance and the erosion of privacy in communal settings where individuals have a reasonable expectation of anonymity [1]. The absence of face detection and individual tracking reduces but does not eliminate the ethical implications, since crowd images captured for density assessment inevitably contain identifiable individuals whose presence is recorded without explicit consent. The risk of demographic bias is genuine given the geographic concentration of training data, and systematic bias auditing comparing model accuracy across diverse crowd types should be performed before deployment beyond the specific geographic and event types represented in training. False positive alerts at Medium density pose a social risk of unnecessary security interventions at lawful public gatherings, potentially creating confrontational situations that themselves escalate crowd tension. The absolute requirement for human-in-the-loop review before any physical intervention is both a technical necessity given the model’s 17–19% error rate and an ethical requirement for responsible deployment. Development and deployment of AI systems for public safety monitoring must be conducted under robust governance frameworks balancing the genuine life-safety benefit of early crowd detection against the civil liberties and democratic rights of individuals subject to surveillance.

1145 5.9. Limitations

1146 The primary technical limitation is the use of static images, which does
1147 not capture the temporal dynamics of crowd development including den-
1148 sity surges, directional flow, and rate-of-change between density levels that
1149 are critical precursors to crowd crush events in real-world monitoring. The
1150 merged dataset of 7,558 images, while larger than single-source crowd stud-
1151 ies, remains modest relative to the visual diversity of real-world crowd sce-
1152 narios across different cultural events, geographic locations, indoor versus
1153 outdoor venues, and diverse camera hardware. All three models were eval-
1154 uated on a single random train/test split without repeated experiments or
1155 cross-validation, limiting the statistical reliability of performance compar-
1156 isons between models whose accuracy differs by less than 2 percentage points.
1157 Although Grad-CAM provides valuable qualitative insight into the visual re-
1158 gions influencing model predictions, the explainability analysis was limited
1159 to a single interpretation technique. Incorporating additional explainabil-
1160 ity methods could provide complementary perspectives on model behaviour
1161 and further strengthen confidence in the interpretation of predictions across
1162 diverse crowd scenarios.

1163 5.10. Future Research Directions

1164 The highest-priority future extension is SHAP-based explainability us-
1165 ing DeepSHAP, enabling cross-validation of Grad-CAM attention findings
1166 and providing pixel-level contribution maps for richer interpretability evi-
1167 dence. Extension to video-based temporal crowd analysis incorporating spa-
1168 tiotemporal CNNs or recurrent architectures that model density dynamics
1169 across consecutive frames would substantially improve practical utility for
1170 continuous crowd safety monitoring. External validation on geographically
1171 and demographically diverse crowd datasets would strengthen generalizabil-
1172 ity claims and identify systematic performance gaps requiring targeted data
1173 collection. Model compression through quantization-aware training, weight
1174 pruning, and knowledge distillation should be investigated to enable real-time
1175 inference on embedded edge hardware without GPU acceleration. Statisti-
1176 cal comparison using repeated experiments with different random seeds and
1177 bootstrap confidence intervals would provide stronger evidence for the narrow
1178 performance differences observed between architectures. Collaboration with
1179 crowd safety professionals would validate whether the classification thresh-
1180 olds correspond to operationally meaningful safety boundaries, and whether

1181 Grad-CAM attention patterns are interpreted as trustworthy by practitioners
1182 who would use the system.

1183 6. Conclusion

1184 This study developed and evaluated an explainable CNN-based frame-
1185 work for automated crowd density classification from static images, address-
1186 ing the critical public safety problem of identifying dangerous crowd condi-
1187 tions before they escalate to life-threatening events. A merged multi-source
1188 dataset of 7,558 images from ShanghaiTech, JHU-Crowd++, and Abnormal
1189 High-Density Crowds was assembled into a balanced four-class corpus cov-
1190 ering Low, Medium, High, and Very High density levels, providing greater
1191 training diversity than any single available crowd dataset. Three CNN-based
1192 models were implemented and compared under identical experimental con-
1193 ditions: a custom CNN baseline trained from scratch, and ResNet50 and Ef-
1194 ficientNetB0 fine-tuned using two-phase transfer learning with class-weight
1195 balancing, augmentation, and label smoothing. EfficientNetB0 emerged as
1196 the best-performing model, achieving 83.27% test accuracy at 3.84 ms per
1197 image, followed by the custom CNN at 82.74% and ResNet50 at 81.48%, all
1198 within a narrow 1.79 percentage point range. The most important practical
1199 finding is that all three models achieve perfect Very High density classifica-
1200 tion ($F1 = 1.00$), confirming that the most safety-critical crowd condition
1201 is reliably detectable regardless of architecture choice, and supporting the
1202 deployment of automated Very High density alerting as a high-confidence
1203 component of tiered crowd safety systems. Grad-CAM analysis provided
1204 qualitative evidence that the custom CNN focuses on semantically mean-
1205 ingful crowd regions for correct predictions, with heatmap attention drifting
1206 to background elements for misclassified samples at the Medium and High
1207 density boundaries. The primary limitations of this study are the use of
1208 static images, which do not capture the temporal dynamics of crowd evolu-
1209 tion, and the absence of external validation on independent crowd datasets.
1210 Although Grad-CAM provided valuable qualitative insight into model be-
1211 haviour, further evaluation using additional datasets and robustness testing
1212 under real-world surveillance conditions would strengthen confidence in the
1213 proposed framework. Future research should extend the proposed framework
1214 to video-based crowd analysis, incorporate external validation on more di-
1215 verse datasets, and investigate additional Explainable Artificial Intelligence
1216 techniques, including SHAP, to complement the Grad-CAM analysis and

1217 further enhance model interpretability and practical deployment.

1218 **Declarations**

1219 The authors used the generative AI tool ChatGPT (OpenAI, San Fran-
1220 cisco, CA, USA) to improve the language and clarity of the manuscript. The
1221 authors reviewed and edited all content generated by the tool and take full
1222 responsibility for the final version of the manuscript.

1223 **References**

- 1224 [1] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, C. Xu, MSA-CNN: A multi-
1225 scale attention convolutional neural network for crowd counting, *IEEE*
1226 *Transactions on Multimedia* 24 (2022) 3967–3977.
- 1227 [2] J. Drury, D. Novelli, C. Stott, The psychology of crowd behaviour in
1228 emergency evacuations: from quiescence to panic, *Current Opinion in*
1229 *Psychology* 47 (2022) 101396.
- 1230 [3] D. Helbing, P. Mukerji, Crowd disasters as systemic failures: analysis of
1231 the Love Parade disaster, *EPJ Data Science* 1 (1) (2012) 1–40.
- 1232 [4] X. Li, W. Zhang, Q. Hu, Z. Zhao, Real-time crowd counting using a
1233 lightweight deep learning model, *IEEE Access* 10 (2022) 86936–86948.
- 1234 [5] V. A. Sindagi, V. M. Patel, A survey of recent advances in CNN-based
1235 single image crowd counting and density estimation, *Pattern Recogni-*
1236 *tion Letters* 107 (2022) 3–16.
- 1237 [6] J. Gao, T. Han, Y. Yuan, Q. Wang, Congested crowd instance local-
1238 ization with ambiguous point supervision, *IEEE Transactions on Image*
1239 *Processing* 31 (2022) 6165–6176.
- 1240 [7] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma,
1241 Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, et al., ImageNet large
1242 scale visual recognition challenge, *International Journal of Computer*
1243 *Vision* 115 (3) (2015) 211–252.
- 1244 [8] X. Jiang, J. Gao, Y. Zhang, W. Lin, Deep learning for real-time crowd
1245 analysis: A survey, *IEEE Transactions on Circuits and Systems for*
1246 *Video Technology* 33 (4) (2023) 1525–1544.

- 1247 [9] J. Zhang, S. Liu, M. Yang, F. Chen, Automatic crowd density classifi-
1248 cation using deep convolutional neural networks, *Sensors* 22 (14) (2022)
1249 5193.
- 1250 [10] M. Tan, Q. Le, EfficientNet: Rethinking model scaling for convolutional
1251 neural networks, in: *Proceedings of the International Conference on*
1252 *Machine Learning*, PMLR, 2019, pp. 6105–6114.
- 1253 [11] W. Hu, L. Zhu, Z. Jiang, H. Fu, G. Li, Blending-target domain adap-
1254 tation for image classification, *IEEE Transactions on Image Processing*
1255 32 (2023) 1525–1538.
- 1256 [12] M. Zhao, J. Zhang, C. Zhang, W. Zhang, A domain-adapted model for
1257 crowd counting in diverse environments, *IEEE Transactions on Neural*
1258 *Networks and Learning Systems* 33 (9) (2022) 4682–4695.
- 1259 [13] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recog-
1260 nition, in: *Proceedings of the IEEE Conference on Computer Vision and*
1261 *Pattern Recognition*, IEEE, 2016, pp. 770–778.
- 1262 [14] W. Liu, Z. Niu, G. Lin, S. Qiu, Uncertainty estimation and sample
1263 selection for crowd counting, *Pattern Recognition* 135 (2023) 109179.
- 1264 [15] D. Lian, J. Li, J. Zheng, W. Luo, S. Gao, Locating and counting heads
1265 in crowds with a depth-aware network, *IEEE Transactions on Pattern*
1266 *Analysis and Machine Intelligence* 45 (3) (2023) 3347–3359.
- 1267 [16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra,
1268 Grad-CAM: Visual explanations from deep networks via gradient-based
1269 localization, in: *Proceedings of the IEEE International Conference on*
1270 *Computer Vision*, IEEE, 2017, pp. 618–626.
- 1271 [17] J. Ma, C. Dai, Y.-P. Tan, Dynamic curriculum learning for crowd count-
1272 ing, *IEEE Transactions on Image Processing* 32 (2023) 2407–2418.
- 1273 [18] Z.-Q. Cheng, J.-H. Li, Q. Dai, X. Wu, A. G. Hauptmann, Rethink-
1274 ing spatial invariance of convolutional networks for crowd counting,
1275 *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44 (11)
1276 (2022) 8049–8065.

- 1277 [19] Y. Zhang, D. Zhou, S. Chen, S. Gao, Y. Ma, Single-image crowd count-
1278 ing via multi-column convolutional neural network, in: Proceedings of
1279 the IEEE Conference on Computer Vision and Pattern Recognition,
1280 IEEE, 2016, pp. 589–597.
- 1281 [20] L. Wen, H. Chang, B. Ma, X. Chen, DENet: Detection-based network
1282 for crowd counting by mapping divided regions, IEEE Transactions on
1283 Image Processing 32 (2023) 3090–3103.
- 1284 [21] F. Zhou, T. Xu, G. Sun, W.-S. Zheng, Learn to count anything:
1285 Reference-guided crowd counting via in-context adaptation, IEEE
1286 Transactions on Circuits and Systems for Video Technology 33 (11)
1287 (2023) 6853–6864.
- 1288 [22] Y. Yuan, X. Rao, Z. Li, Crowd localization and counting based on a
1289 multi-scale feature pyramid network, IEEE Transactions on Circuits and
1290 Systems for Video Technology 33 (5) (2023) 2307–2318.
- 1291 [23] Z. Zhao, J. Li, S. Gao, K. Lu, Domain-general crowd counting in unseen
1292 scenarios, Proceedings of the AAAI Conference on Artificial Intelligence
1293 37 (3) (2023) 3699–3707.
- 1294 [24] Z. Wang, Y. Liu, S. Qiu, C. Bai, Crowd counting via perspective-guided
1295 fractional-dilation convolution, IEEE Transactions on Circuits and Sys-
1296 tems for Video Technology 32 (8) (2022) 5216–5227.
- 1297 [25] C. Xu, D. Liang, Y. Xu, S. Bai, W. Zhan, X. Bai, M. Tomizuka, Au-
1298 toScale: Learning to scale for crowd counting and localization, Interna-
1299 tional Journal of Computer Vision 130 (2) (2022) 405–434.
- 1300 [26] Q. Song, C. Wang, Y. Tai, C. Wang, J. Li, Boosting crowd counting
1301 with partition-based coherent loss and dense feature pyramid, IEEE
1302 Transactions on Image Processing 32 (2023) 1390–1403.
- 1303 [27] H. Lin, Z. Ma, R. Ji, Y. Wang, X. Hong, Boosting crowd counting
1304 performance by fusing multi-scale features through a learnable weight
1305 mechanism, IEEE Transactions on Multimedia 25 (2023) 4911–4922.
- 1306 [28] J. Chen, B. Su, Z. Wang, Efficient crowd counting via structured knowl-
1307 edge transfer, ACM Transactions on Multimedia Computing, Commu-
1308 nications, and Applications 19 (2) (2023) 1–22.

- 1309 [29] Y. Tian, L. Pan, F. Maire, J. Barthelemy, Counting crowds with limited
1310 labelling through a semi-supervised learning approach, IEEE Transac-
1311 tions on Intelligent Transportation Systems 24 (3) (2023) 3714–3725.
- 1312 [30] Y. Abouelnaga, H. M. Eraqi, M. N. Moustafa, Real-world anomaly de-
1313 tection in surveillance videos, IEEE Transactions on Circuits and Sys-
1314 tems for Video Technology 32 (7) (2022) 4408–4418.
- 1315 [31] J. Wan, Z. Liu, A. B. Chan, Represent, compare, and learn: A similarity-
1316 aware framework for class-agnostic counting, IEEE Transactions on Pat-
1317 tern Analysis and Machine Intelligence 45 (2) (2022) 1418–1434.